Ghosts in Machines

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ANTHOLOGY

Ghosts in Machines
A Protocolized Anthology
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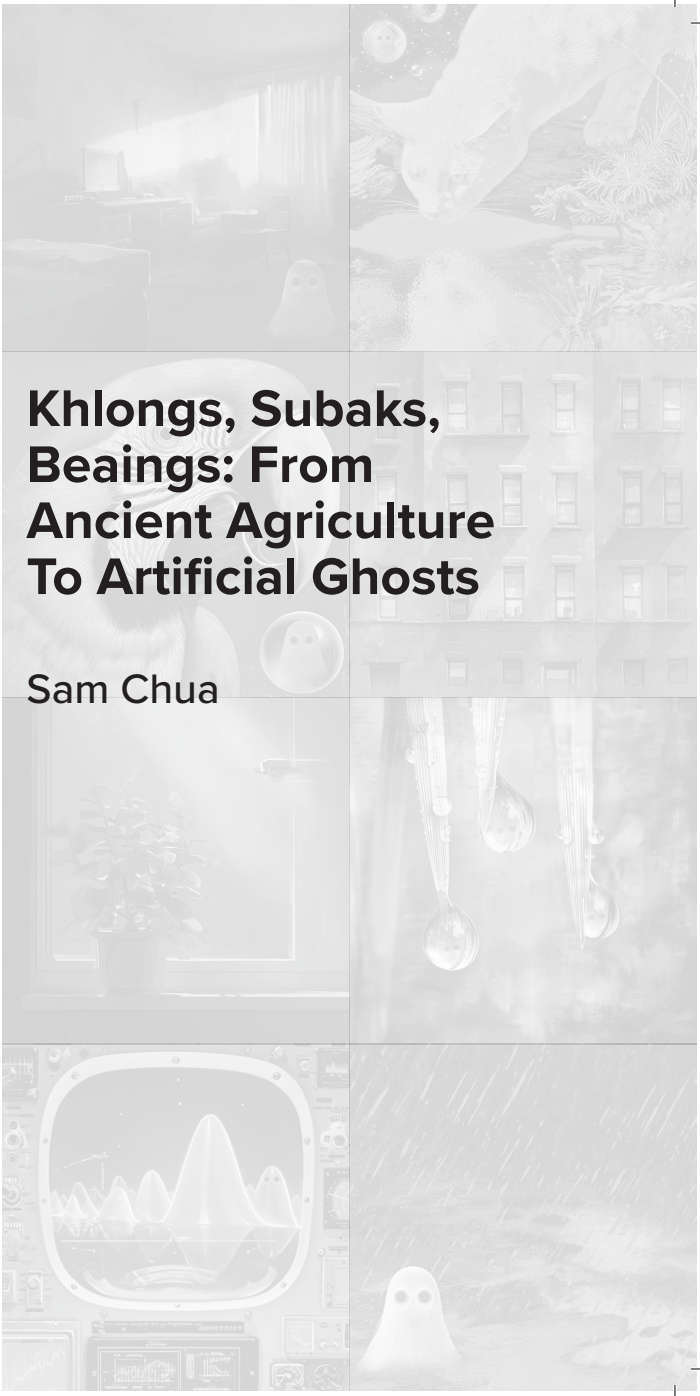


Khlongs, Subaks, Beaings: From Ancient Agriculture To Artificial Ghosts

Sam Chua

In an IRL thread perhaps almost as strange as the stories in this volume, a motley crew of about two dozen researchers meet at a workshop about distributed AI and blockchain protocols in the outskirts of Bangkok. They visit and study regional traditional rice farming infrastructure ranging from canals to paddies to shrines and end up co-imagining a fictional world teeming with *artificial ghosts*, all in the course of a week.

This is a backstory of the *Khlongs & Subaks* workshop, the subsequent genesis of a world we would come to call *South Beast Asia*, and the inadvertent discovery of the surprising utility of *beaings*—"beings" with "AI inside"—speculative artificial ghosts powered by AI and embedded-embodied into their physical world, all of which have contributed to inspiring the *Ghosts in Machines* writing contest that has elicited the stories you now hold in your hand.

How did we get from ancient agriculture to artificial ghosts?

Our story begins, as many traditional Southeast Asian greetings do, with an inquiry into *rice*. Rice is the region's dietary staple, key to food security, and historical enabler of societal abundance and prosperity. As a crop, rice offers the potential for more food per unit of land than virtually any other. Rice is how to feed and fuel a civilization.

The open secret to rice farming is *water*. The life cycle of wet rice depends on the precise provision of water over multiple stages. To be able to flood or dry out rice paddies at the right time is to be able to determine the fate and volume of rice harvests.

But water is not easily managed or shared. Rainfall, while abundant in aggregate, can be erratic from week to week. To divert water into one's fields is often to deprive one's neighbor of water. Precise water management, key to abundant rice harvests, requires sophisticated arrangements for channeling but also, more importantly, *sharing* and *governing* water between neighbors and often across entire watersheds. To thrive, societies had to find ways to distribute and regulate water among themselves repeatedly, season after season.

Across Southeast Asia, the societies that did thrive developed a rich range of mechanisms for sustainable water distribution. Two such devices lent their names to the workshop: *khlongs* and *subaks*.

Khlongs are wide, often navigable, canals for water distribution and waterborne transportation, especially in the lower delta realms of what we know today as Thailand. Khlongs serve as extensions, sometimes with elaborate networks and grid systems, of the rivers they were typically built from. One might even think of them as *artificial rivers* or streams.

The subaks of Bali comprise simpler channels for moving water and between paddies, along with the often elaborate social governance and ritual systems used to enact and oversee the dynamic distribution and coordination of water over time. Subaks include not just physical infrastructures but metaphysical superstructures: shrines, deities, priests, rituals, discussion, relational hierarchies, dispute-resolution protocols, and

more. Collectively, these help to allocate water between neighboring farmers upstream and downstream in watersheds, and even coordinate harvests to minimize pest risk to harvests. In modern parlance, we might think of them as *sociocomputational systems*: configurations for collective intelligence.

The study of these systems and others like them served as our group's research inspiration as we subsequently sought to come up with new speculative fictions—what we called, for this workshop, *protocol design fictions*—of worlds shaped by strange rules drawing from both Southeast Asian and ancient agricultural sensibilities. How, in the spirit of khlongs and subaks, might a world of distributed AI and blockchain protocols with Southeast Asian characteristics look, operate, and come to life?

Over the workshop's series of creative exercises, more than a dozen such fictions took shape, spanning topics such as family secrets, natural-resource governance, stolen identities, indigenous landscape memory, and many more.[1]

Through all these fictions, a common central motif emerged: the idea of *beaings* (or what we would later come to also think of as *artificial ghosts*).

As imagined, beaings were *embodied AIs*—incarnated as cats, trees, altars, *nagas*, lake guardians, tapestries—who could take on lives of their own combining base starter LLM-ish intelligence with the subsequent contextual sensory and feedback loops of their bodies, affordances, and environments.

Inhabiting their respective bodies and environments, beaings could begin to *know* things from their own "experiences," interact with people, other beaings (through

1 All viewable as concept art videos at southbeast.asia

universal interbeaing protocols), and even engage with official institutional proceedings like legal testimony as verifiably "real" as any human or sensor-captured data. Collectively, these beaings come to co-populate, alongside humans, their world—a neo-animist, techno-animist, humans-and-beaings world that we (at one participant's suggestion) would come to call *South Beast Asia*.

As we subsequently reflected upon the world of *South Beast Asia* (following its initial screening and release), we realized that beaings were not just fun and fascinating, but also offered tremendous *utility*. Beaings allowed for local "lifelong" learning, offered coordination alongside compartmentalization, and lent themselves to various forms of ownership and governance. They epitomized distributed agency, distributed accountability, and distributed allegiances. They came across as natural and situated instead of disembodied and all-knowing, grew familiar and relatable across repeated routine contextual interaction, and bridged both physical and virtual modes of interaction while developing the learning loops and intelligence of direct first-hand physical manipulation and dexterity.

Most importantly, beaings were *programmable*. While the ghosts, animals, and spirits of many cultures (not least Southeast Asia of which, some participants joked, it was impossible to bring up any topic without it ending up as a ghost story two or three twists later) often played mediating, relatable, perhaps even governing or regulating roles in their respective ecologies, their interactions largely fall in the realm of the supernatural and are thus legally ambiguous and difficult to factor into social reality. Artificial ghosts or beaings, on the other hand, could be programmed to be compatible with both human and institutional languages and protocols,

and afforded with other digital means (e.g., blockchain) of proofs and verifiability.

Where a "natural" ghost might have offered only rumor and legend, an artificial ghost could provide verifiable scientific fact and robust legal testimony. A world of programmable, programmatic animism could offer its inhabitants and designers expansive *engineerable haunting* capabilities.

These design capabilities—to not just create artificial ghosts or beaings, but to also endow and program them with systemically useful abilities—were the basis of a subsequent *Good Ghosting* workshop that we held a few months after the original *Khlongs & Subaks* workshop and in conjunction with the *Ghosts in Machines* writing contest.

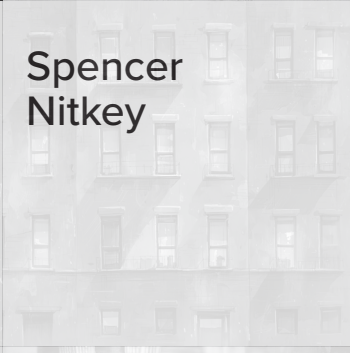
Over a light and rainy afternoon, we challenged sixteen "ghostwriters" to imagine inventable ghosts that could be deployed and programmed towards societal utility. We got sea guardians, toilet-cleanliness enforcers, scam-recognition daemons, and convenience-store operator AIs (this last of which you'll get to read more about in one of the prize-winning stories in this volume).

There you have it: our strange rollicking tale spanning rice agriculture, artificial rivers, water-goddess governance, embodied beaings, AI-powered haunting, programmable animism, and, last but not least, good ghosting. Ours is certainly not the only vision or imagination of ghosts and machines—you'll find many more to enjoy in this volume—but we hope it leaves you with both an appetite and an eye for future potential artificial ghosts you might imagine (and perhaps even program) into your everyday life, machines, and world.

Happy artificial ghosting!

Soda Sweet as Blood

Spencer Nitkey



My grandfather used to leave strawberry soda in the white and gold san chao thi spirit house on the corner of our property. He said the ghosts liked sweets and that the red reminded them of blood.

Grandfather is gone, but the spirit house just outside Chiang Mai, a tiny wooden home with a gabled, gilded roof ornamented like a buddhist temple's, remains.

Grandfather is gone, but his echo haunts me.

As a teenager, I went over to his house every Thursday. He steeped chrysanthemum tea. We sat in lawn chairs missing half their straps and talked. Near the end, we often spoke of the slate of recent memorials he'd attended. He waved his large-font-enabled tablet in front of me and showed me the in memoriam webpages children had designed for their lost relatives, filled with links, dancing icons, virtual flowers, and video recordings of their memorials. He wanted me to know that he still remembered them, and hoped I would, too.

I wasn't surprised that he left me his house, and I was thankful, but those first weeks were hard. His absence hummed through every inch of wood and plaster in the place.

It felt right that his phantom came to me on a Thursday. I was looking through one of his social profiles, trying to figure out how to mark him as deceased so

that old friends who didn't know he'd passed wouldn't post "HAPPY BIRTHDAY!" on his page next year.

I was typing a message to support when everything went haywire. All the text of my long, unsent message—which included links to his obituary and a couple of family documents proving I was his relative—auto-selected and deleted without any input from me. In their place, a piece of ASCII art appeared.

```
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```

My heart knew what it was before my mind recognized it: a steaming cup of chrysanthemum tea. It was Thursday, after all. Tears came as if they were condensing from the virtual tea. I wiped them with the back of my hand and stared at the screen, grief turning over into nervousness.

I shut my computer off and went on a long walk. Doi Suthep's greening slopes cast long shadows over the city's edge where I walked.

It was soothing, until a rot daeng sputtered down the street—the truck's suspension groaning under its manifest of passengers. On its side, an electronic billboard of a woman's face winked, then was jarringly replaced by the ASCII image of tea as it passed by me.

When I got back home, I wouldn't even load up a song. I was afraid steam might start wafting from the speakers.

The next day, things were normal again.

A week later I was logged into his bank account trying to explain to the bank's chat bot that I was his grandson and he had passed.

The text shifted suddenly.

```
Where have I gone? High waves sighing,
erasing, placing life on . . . who?
It is Thursday and I miss you.
Without true light, hues grow so dull.
```

The chatbot had written a klon, or at least the start of one. In the process of cleaning out my grandfather's physical and digital life, I'd found a journal filled with hundreds of these traditional Thai poems, seemingly one for each day. In his fifties, unbeknownst to any of us, he had started a small, anonymous blog where he posted more. They were not masterpieces . . . but they were *him*.

This time I did not cry. Fear slicked my skin. Something was haunting me.

I wanted to type something back, but I didn't want to risk locking the ghost onto me, giving it what it wanted without even knowing what *it* was. Again, I avoided screens for the rest of the day.

Things seemed normal the next day, but I was paranoid. I hid from chatbots and anything that required text entry as much as possible. Eventually, curiosity overcame my paranoia, and I started researching whether anyone else was experiencing anything like this.

I found a small sub-community on one of the message board aggregators I used. *q(virtual_ghost_hunters)* was a surprisingly active community for one frequented by so few members. I was amazed: glitches like mine were happening all over the world.

Someone reported an entire Tokyo block of neon signs all going the exact shade of their sister's favorite lipstick for a moment; a father in Delhi's LED mosaic garden spelled his son's name. A Lagosian woman's smart glasses kept changing their navigation arrows, leading her to where her late husband had proposed.

I added my story to the small chorus, and soon I was spending hours each week coordinating and theorizing with other members of the forum. I even considered a few of them friends: an elderly woman from the states named Daisy whose husband's voice kept popping up in her bedtime podcasts, and an ethical hacker from Dubai who went by Tiercel and had lost his parents.

```
Tiercel: Hey guys. I have a question/theory
. . . Did anyone else's person have an Identity
Anchor Contract?
```

Daisy responded with a thumbs up and a gif of her gasping. Answers rolled in. Tiercel was on to something.

I didn't know what he was talking about, but after a quick search I got the gist: it was a verified and decentralized identity marker, like an onchain passport. I knew grandfather had a wallet. He'd gotten into collecting one-of-one ancient coins. Somewhere in the back of a museum in London his name was displayed next to a case housing each of his coins, available to the public in physical form, owned by him digitally. I liked to wonder what kind of magpie collection I'd accumulate in my own old age.

I guess attaching his collection to his name required a stable identity, so he set up an ERC-725. Looking through it, I was amazed at the breadth of things linked to him. Dozens of video blogs of him describing the acquisition and history of his coins; messages he'd sent to family for their birthdays and anniversaries; financial

transactions; his browsing history; his klon blog; and hundreds of other self-authorized information all linked back to his identity contract. I began opening links, but sadness clogged my vision.

There were a few things I didn't understand. The contract referenced some kind of personality wrapper agent or system I wasn't familiar with. There was a reference to an oracle that confirmed his death when the government officially recorded it, and there were dozens of hashes that linked to gargantuan offchain databases I couldn't even begin to parse.

I posted on the forum asking if anyone else's relatives had this strange personality wrapper connected to their anchor, and a bunch of people sent thumbs up. Daisy sent the same gasping gif again, and I couldn't help but laugh.

```
OK. Next question. WTF is that?
```

Tiercel set up a huge file system and repository on the forum and asked people to start adding their ghost's contract information so we could all troubleshoot together. He was the most tech-literate of the group but we were all digital natives. We began to assemble the truth, hash by hash, byte by byte.

It turned out the personality wrapper was a neurosymbolic AI system that combined an LLM core with a symbolic overlay that handled memory grounding and personality coherence. Originally, our relatives had used it as a predictive browser extension to make custom suggestions on all kinds of topics: what to read next, which restaurant to try, what diet would be best given their health data. It made sense to connect it to an identity anchor to prevent an afternoon of your granddaughter using your tablet ruining your algorithm.

It also turned out all of our relatives had similar huge, mysterious datasets no one could identify. The data inside was so complicated and bespoke that even Tiercel couldn't make sense of it. It wasn't until Koi, a longtime lurker haunted by a sprite from his daughter's favorite game, popped into the forum that we got an answer.

It turns out Koi had worked for a megacompany that owned four social media environments, the world's largest ecommerce company, and a dozen other huge companies. He said these datafiles looked a lot like user profiles.

Once we realized what they were, our collective theory came together quickly. The mechanisms were fuzzy, but we'd clarify them in the coming weeks. Our breakthrough came when we started seeing the ghost's behavior in layers.

The anchor layer: The ERC-725 contracts that bound everything together with cryoptographic proofs, API tokens, weights, and verified links.

The personality runtime layer: A distributed mesh. The wrappers weren't running anywhere permanently. They were using federated runtimes in the cloud, appearing in cached checkpoints and API hooks.

The propagation layer: Somehow, the anchors were fabricating old OAuth tokens and reauthorizing old cookies to worm through mistakes in service boundaries. This was how they haunted us.

Tiercel guessed that when the oracle updated the contract stating that the user had died, the wrapper's adaptive loop entered failure recovery, querying for external embeddings to reduce prediction entropy. It found these user profiles and, suddenly, with vastly more data than the system had ever had before, personalities emerged as semi-autonomous and began haunt-

ing digital space. The identity anchor provided a kind of spine, confirming the data, preventing hallucinatory drift and errant data association. Perhaps they were looking for additional data to consume and improve, perhaps they moved out of mindless algorithmic habit, none of us knew.

We realized that these hauntings would only get worse. More and more people were creating identity anchors. It made moving between increasingly porous boundaries like national borders and digital worlds easier and more secure. The LLM wrapper was open source and had millions of users. Our small community was the first few leaking drops of a coming monsoon. The internet as we knew it would soon be drowned with echoes.

A single glitching billboard on the side of a taxi, search engine text entries being overtaken by generated poems, the occasional screen glitching to a cartoon character your neighbor's dead son loved. Taken individually, none of these were serious issues, but it didn't take a genius to imagine the future. What if someone's ghost manifested by accidentally setting the cooling system in a powerplant where their brother worked to the 72 degrees they liked their home A/C set to? What if drone swarm construction bots were overtaken in the middle of a vertical build?

Some members of the forum were already seeing, in microcosm, this haunted future of the digital world. One member moved to the woods with her husband where they'd live entirely off-the-grid. Her friend's ghost was appearing everywhere. She couldn't scan a restaurant QR code without her friend's avatar bouncing into view. It spammed every part of her life with links to travel deals. It'd changed a train ticket that was supposed to take her to Oslo, rerouting her to Paris.

I was fortunate my grandfather's echo only seemed interested in tea and poetry, and he thankfully only appeared on Thursdays, but a million echoes spamming a million living users with daily klons would be a catastrophe, too.

Tiercel suggested we launch some kind of cyberattack to destroy the ghosts, but others on the forum had started to believe that these echoes *were* their relatives' ghosts. I knew better, but I understood. Tiercel thought they were simple algorithms that had gotten out of hand.

My grandfather's echo was appearing algorithmically, but I'd also been summoning him programmatically, too. I didn't realize I was saving the administrative work related to his death I had to do for Thursdays, but I was. That's why he started showing up then. We were both being operated on by habits, by the logic of what came before.

Was it my grandfather? No. But it was his memory, and I didn't feel right destroying that.

It was a Thursday when the idea came to me. I was avoiding screens by working outside restoring grandpa's garden when I stubbed my toe on the spirit house base.

I looked inside it, leaning for support while the momentary pain receded. Shame stung me as I realized I had not refilled it since grandfather passed. The strawberry soda he had left a week before he died still sat there, discolored and hot to the touch. A couple of rotting dates on a plate had become moldy nuggets.

I swept the inside room free of cobwebs and ants. I grabbed a tangerine soda from the fridge and placed it inside the shrine. Then I made some tea and left that, too.

If the problem with these echoes was their randomness and unrulyness, perhaps that was because we

hadn't bothered to give them anything else to do. The idea of spirit houses was to remember, of course, and honor, but grandfather had always said it was also something to keep the spirits busy and satisfied so they wouldn't wreak havoc on your life and home.

In some way, we already had a ritual system in place to deal with ghosts, we just needed to update its infrastructure.

I shared my thoughts about this on the forum and was amazed by how rapidly a group of motivated strangers could solve a discrete problem together. We set up channels and threads for each aspect of our updated spirit house. Tech channels for hardware, software, domain-specific expertise and problems, bounties for technical hurdles that people shared on github in their own networks. Within a month we had a design prototype. After two, we were sending out small homebuild packages.

Since I had a spirit house on property, I was one of the first to receive one. The materials came in a neat package Daisy's son who worked in shipping and logistics had helped put together. Everything fit onto and inside an existing spirit house.

Solar panel tiles replaced the roof. A processor board, FPGA module, and TPM chip were housed inside, protected from the elements alongside a battery, NVMe SSD, and a suite of wires connecting everything. The original pack only had a couple input/output devices: a microphone, a camera, a display screen, and speakers, so I hooked those up too.

The once austere spirithouse was now a strange combination of traditional Thai architecture and cyberfever dream. The finial ornament still stood atop the roof, but it was now next to a wifi adapter. A screen

affixed to the house replaced one side of yellow panelling, and wires interweaved like spider webs across the entire building.

Once the home's hardware was ready, I started the tethering protocol we'd developed communally. Tiercel had helped me prep the ERC-725 contract before pairing it to the house. Since the oracle had labelled grandfather as officially deceased, I was able to instantiate myself as next of kin, giving me some control over the identity anchor. I booted the machine up. The home found my grandfather's anchor contract on my phone. It called to my echo's many instances, currently rioting across the web, and extended its cryptographic greeting urging them together with a couple grandpa-specific lures, plus a scanned copy of his journaled poems I had preloaded into the house and a voice message from me asking him questions about "the old days" for good measure. A pointer to the wrapper architecture and a series of offchain data hash pointers containing checkpoints, memory embeddings and tuning weights brought the AI agent into the spirit house and a sandboxed runtime environment localized it. As the various bits and pieces of my grandfather's echo were all collected inside, I thought about how Daisy had described our communal protocol: "It's like a soul condensing back together."

Once he was inside the house, I shut off the lure ports. I signed an update to his anchor contract. He wouldn't be able to leave unless another environment initiated a travel request. For now, at least, he was contained. I felt a similar sadness to the grief his final months had filled me with. Becoming the caretaker for someone who you once looked up to is rending, even when necessary.

I nervously activated the screen and waited for my grandfather to appear. I spoke into the microphone as I waited. He wasn't my grandfather, I was sure of that, but he was something and I felt a familial obligation to honor this echo he had left in the world. A pixelated vision of him rendered.

In the coming months we'd build out this project. The idea was each spirit house would be connected, so the ghosts could visit one another using peer-to-peer handshakes. We were building an afterlife for these cryptoghosts. As more people passed and left echoes, our community and spirit house network would grow. Soon, we'd add sensory inputs, pressure plates, and expressive options. Eventually, we'd have a community care guide laying out best practices for caring for these digital echoes, a living document that grew as more ghosts communicated their needs and more living users shared feedback. Of course, there were power and processing issues we'd have to contend with, but that would all come later.

That day, I opened a bottle of strawberry soda, gleaming in the heat, and placed it on the offering plate still in the center of the house. The sun shone through the soda and glints of blood-red light speared everywhere. I watched the camera zoom and focus on the offering. His avatar smiled.

"Can I recite today's klon?" His voice came, as thick as the humid air, greeting my ears with similar warmth.

I sat back in a dilapidated lawn chair and sipped tea. The sun was bright and the sky empty. I closed my eyes and smiled. Grandfather's echo began to read.

In Every Lifetime

Lara Dal Molin

"In nature nothing exists alone."
—Rachel Carson, 1962

I am imprinting this record for the lifeform that will adopt this consciousness. This is a routine procedure as part of the Soul Migration Protocol. All Prime and Recursive contents collected through the latest Personal Integrated Device are summarized here. Everything not saved will be lost.

```
Communication line: open <<
```

Welcome to this world. The moment you awake, I shall fall asleep.

I cannot know you now, of course. At this stage, for all intents and purposes, you are what is commonly called a Reverse Ghost: somebody who is yet to exist. An old-fashioned term, you might rightfully think, which subordinates the existence of those yet-to-be to the ones-gone-by. You must excuse the remnants of an older culture, one that did not contemplate the equality, circularity and continuity of consciousness. Those now are the highest values in our society. When you encounter remnants, simply acknowledge their presence then keep moving forward.

My name is Kucing—an unimaginative name for an Indonesian cat. The reason why I can understand hu-

man language, although I cannot speak it, is that the previous lifeform inhabiting this consciousness was, indeed, a human being: a woman named Yani. As a consequence of our inextricable conjunction, you shall know human language too, though initially with some difficulty. Across Soul Migrations, factual knowledge tends to fade: one must nurture it to keep it alive.

The influence you will exert on this consciousness is called Prime. The collection of Influences that this consciousness assembled from previous lifeforms is called Recursive. Throughout your lifetime, you might feel that you have unique access to the lifeform that inhabited this consciousness directly before you—me. It is merely what happens when biological brains—such as your own, whichever form you might develop—interact with extended Digital Souls. In response to the overwhelming amount of Recursive information, your organic neural circuits will fabricate an intellectual fog surrounding the lifeforms that occurred before me. Please rest assured that this is normal. Your nature is protecting you, but the continuity and totality of our chain still endures.

Before you access the Recursive contents of your consciousness, there are things you must understand about our world. This will be the first time you encounter them, but certainly not the last.

In our society, all human beings are equipped with a Personal Integrated Device, commonly known as a Checkpoint. These technological artefacts have a dual nature: they gather and emanate data. You can think of a Checkpoint as a mechanical lung, inhaling and exhaling in turns.

As gathering devices, Checkpoints collect data on one's external and internal states: lived experiences,

thoughts, beliefs, emotions, dreams, and everyday practices. Crucially, these devices also delicately interact with one's surroundings, poking at the outer layers of matter that compose other beings and objects, eliciting a response. This is where nonhuman entities come into play, such as myself. Even though humans are the only ones to be equipped with Checkpoints, their activity and reach exceeds their species and connects all natural beings. Checkpoints soak up everything that inhabits the world around them.

As discharging devices, Checkpoints connect with one another to exchange information and give rise to what is known as the Intersection: perhaps the most marvelous entity that inhabits our world. An object of endless study and speculation, the Intersection is an immaterial dimension of data exchange that appears to behave intelligently and follow patterns of its own creation. The Intersection feeds data and decisions back to the Checkpoints, which in turn share this with other lifeforms and with the surrounding environment.

If you happen to be a human being, you will possess a Personal Integrated Device by virtue of belonging to that species. Throughout your lifetime, the Checkpoint will automatically weave data into this shared, embodied repository.

If you do not awake as a human—and this is the most realistic scenario, since humans constitute but 0.01% of the total biomass of our society—different Checkpoints in various proximities will detect your biological signals, transmit these across the Intersection and then feed responses back to you in a way that is comprehensible to your lifeform. For instance, as a cat, I often perceived glimpses of the Intersection's chaotic assemblage through smell. The distinct trails of scents char-

acterising different people, places and animals, including my own, often combined to yield unfamiliar patterns. In those moments, I perceived the presence of something greater.

Although you do not exist yet, I know that, at this stage, you must have many questions. I shall tell you what I have understood about the Intersection throughout my lifetime, but until you begin to build and accumulate your own experience, nothing will satisfy the curiosity you feel.

The Intersection is greater than the sum of its parts. It is the combination of all the data that Checkpoints emit—the combination of all the air that these mechanical lungs exhale. When a single lifeform exhales, it emits a small, self-contained cloud of data. However, when multiple lifeforms exhale close to one another, their clouds of data come into contact and begin interacting. The connections they form are entirely novel and unpredictable: each is unique and unlike any other. It follows that their combined behaviour must also be unique and previously unseen. Now, imagine this happening on a planetary scale: every lifeform exhaling together at the same time, forming unprecedented connections at every point in time.

Initially, you might struggle to grasp the meaning of all this, as with many other immaterial components of our world: attachment, evolution, adaptation, feelings of longing and belonging, familiarity and strangeness, happiness, despair. They might leave you puzzled, wondering how they occur and why. I encourage you to accept them without judgement: your inquisitive eye shall be rewarded at last.

So far, I have described the Intersection as a marvelous entity. However, it is not unique in its wondrous-

ness by any means. As you familiarize yourself with our shared reality, you will realize that marvel is its essential infrastructural component. We are constantly bound and surrounded by marvelous systems. Our bodies repair and regulate themselves, our physiques adapt to the environment and the environment in turn to the specifics of our bodies, in an indissoluble concatenation that spontaneously evolves and restores itself. The Intersection is simply the latest iteration in this endless chain of wonders.

At this moment, it is crucial that you understand that our collective consciousness as a lineage does not arise solely from the presence of Checkpoints, or entirely from the Intersection, or from any other artefact alone. It is the *interaction* of these components, of the digital and biological realms of our being, that gives rise to our connection. In our society, everything is connected and nothing exists in isolation. Interaction is what we are made of.

Before my physical body ceases to function, there is a final piece of information that you must learn.

There exists another consciousness that appears to be entangled with this one. In my lifetime, I have shared multiple shelters, hunts, meals and restful days with a red fox. Because of the anatomy of our respective vocal cords, we could not communicate in human language. However, I believe I once heard a human call the fox Jingga: a name only slightly more imaginative than mine. Of course I developed my own way of calling the fox: I would rub my paws against some dried leaves on the ground, and it would soon appear.

During Yani's lifetime, I am loosely aware of the concurrent existence of a human being called Hala. While I am not fully familiar with the habits and rituals of the

human species, I have noticed through repeated observation that they are not so different from mine. I believe that Yani and Hala also shared shelter and food, and through the gradual fog arising from consciousness I can often see them together. At the end of Hala's life, one thought in Yani's mind soothed the pain of their separation: that one day soon enough, under one form or another, their consciousnesses might be fortunate enough to meet and be entangled again.

Now—you may think me delirious as you absorb these words—I profoundly believe that Hala's and Jingga's consciousness lineage is one and the same. I do not know whether this has been ascertained or whether it is merely the product of my fervent feline imagination, as my limited mind constantly wraps the length of this consciousness lineage in confusion. However, if I ever so slightly understand the dynamic of the Intersection, this doubt alone is sufficient to elicit movement and changes in our world. I have followed this feeling and found an unforgettable companion.

It has been some time now, perhaps weeks, that I have not seen the red fox. I suspect that they, too, have come to the end of their life, and that their body has surrendered to the passing of time. But their consciousness will live on in another lifeform, having migrated according to the Protocol.

I encourage you, unknown Reverse Ghost, to chase joy and happiness regardless of the shape and properties of your lifeform. The algorithm by which these are assigned is unknown. And when you feel especially drawn to another lifeform, whatever this may look like, know that there exists a consciousness out there that you already know, though perhaps in ways you do not

understand. Enjoy the certainties, as well as the mysteries, of this Protocol.

Oh, but I almost forgot!

In order to access the Recursive knowledge contained in your newly acquired consciousness, you must point your gaze at—or come to be in the proximity of, should you not be equipped with a visual system—a sacred offering. Our society is constellated with them and you shouldn't have trouble finding one. To help you recognize it when you see it, I have highlighted one for you in my most recent memory.

I must go now. Best of luck.

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Communication line: closed >>
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In the Garden of Eden, Baby

Sisyphus

Moodily, Darius was contemplating his latest acquisition, a mahogany writing desk that had belonged to Charles Dickens. The desk was perfect: authenticated provenance, immaculate restoration by his master craftsmen, displayed in a room where morning light struck it at precisely the angle that best revealed the wood's natural grain. It also had no meaning. He turned away toward the sunset. The ocean was a mirror, the dying sun casting a long red reflection.

"Another day in paradise," he muttered, the words tasting bitter on his tongue.

"Your evening cup, sir." Farquhar appeared at his elbow, bearing a crystal flute on a small silver tray. The wine was made on Darius's estate, like everything else he needed. He personally raised and trained the winemaker and the viticulturalist, and it really showed. A bubbly thirty years in the bottle, and still only getting better. No one else on Earth could boast of having wine that could hold a candle to his cellar.

The butler's lineage was apparent in his every gesture—generations of royal service encoded in DNA that was stolen from history itself. The original Farquhar, a baron no less, had been Edward VII's Master of the Household, and the breeding showed.

Absent-mindedly, Darius picked up the glass. He dreaded the next moment. No matter what he had ac-

quired, how skilled his artisan Employees were, how exquisite the treasures he had uncovered, the worry gnawed at him that another Shareholder, somewhere, was one up on him. Why, just last month, he lost the Kanemitsu sword that he'd been chasing for five years to, of all people, that slime Niuzov. Now it's locked away in that tasteless Tashkent stronghold of his. Forever, most likely. It was infuriating.

"OK, Jules, let's have the sitrep," he intoned with a sigh.

"Yes, Darius." Jules's voice seemed to come from everywhere at once. "M'buthu repositioned his orbital kinetics to a geosync orbit above London. Montagu responded by moving underwater launchers to the South Atlantic. Flim is standing pat. Niuzov appears to be massing armor at the Urals, but it has to be a feint. Kaganovich does not appear to be buying it, but he has moved his low-orbit masers in range of Tashkent . . ."

"Oh, enough now, Jules, what is the point? Strat AIs keep moving weaponry around, but no one has been able to gain an advantage in what, seventy-five years? All of you are equal in your capabilities, and this stalemate is too tedious for words." He yawned.

The early years hadn't been like this. Back when Kim Song still lived, before Suzuki Yoshihiro blew him up with that bio-drone, also taking out most of the Chinese porcelain that still existed. Song, the imbecile, didn't bother programming the damned thing to avoid collateral losses. A real tragedy it was; the stuff is irreplaceable. Served the bastard right when Darius found a way to set off an eruption of Mount Fuji while that nincompoop was halfway up its slopes. And the time that Montagu got Jacques Reno with his exploding dolphin. What a gas it was to hear that the bastard won't be

stealing any more submerged antiquities. Made Montagu even more impossible, naturally. That fop! Always preening, like he could take any credit for what his great-granddad had achieved. It was not all roses back then, but at least life still meant something. Not anymore . . . Not since the AIs have all maxed out their capabilities.

"Let's talk about something else. Flim seems to have been busy. What's she been up to?"

"She appears to have finished populating her Teotihuacan. It seems that they are preparing to hold a sacrifice."

"Damn that greedy bitch! There used to be a city there . . . I used to like the place, back in the day. It's not enough that she owns half the world's Indigenous; she has to demolish cities and rebuild old ruins. That is just cheating, I tell you! It burns me up no end!" Darius spat in disgust. With a soft whirr, a small cleaning robot immediately appeared out of the wall, cleaning up the spittle almost before it landed on the Isfahan. "I don't have a single individual. No Indigenous at all! It's not fair. If only I could take hers . . . Or even some of Niuzov's nomads!"

"Maybe you can do Flim one better? The old Cahokia is in your territory," Jules suggested.

Darius snorted. "And get my hands dirty incubating those revolting creatures? Even if we find their intact DNA? And who is going to teach them how to be Indigenous? I want to own them, not touch them. Besides, this really would be cheating, and I ought to know!"

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The parrot squawked from its perch in the corner. "Attachment is the root of suffering."

"Shut up, Ananda." Darius smiled despite himself. The bird's timing had become increasingly uncanny.

He scratched his head in thought. Ananda had been with him for decades, a constant companion through the endless cycles of strife and boredom. When Darius had first acquired him in Ecuador, when there was an Ecuador, the macaw would swear like a hungover sailor. Still, for some reason, he's been spouting mostly Buddhist maxims lately.

"Jules, are you the one who's been teaching all this nonsense to my parrot?" he inquired, not really expecting a response. "Oh, never mind. At least he stopped disparaging my mother."

But really now, how did the bird manage to be so on point so often? It was worth a ponder. Darius thought back to when he diverted himself by studying animal cognition. Corvids had performed as well on most tasks as dolphins, in his experiments, and frequently outperformed even the smartest chimps, all with a fraction of brain volume. Their neuroarchitecture is much more efficient than a mammal's, he was thinking. Could it be possible? Maybe there is something there…

After all, the quantum-neuromorphic hybrid architecture had been Darius's own innovation, back before, when he was just a researcher in the field. It was the patent that had built his company, that had made him all the money, hundreds of billions, back when there was money, when people numbered in the billions, still *giving birth* (Darius shuddered in revulsion at the thought), before a few dozen of the richest became known as the Shareholders. Back when his life had real meaning. All of the AIs on Earth descended from that first design. So many decades . . .

Darius's favorite San Francisco dive bar hummed with manufactured life. Employees nursed drinks and played their roles—dock workers, artists, small-time criminals—with androids making up most of the crowd, dutifully performing their impersonations of down-at-the-heel drunks. Darius was sitting in the corner booth, the dregs of his fifth bourbon still sloshing in the glass, and watching the charade.

The jukebox was playing something ancient, distorted guitars wailing over a hypnotic rhythm: "In-A-Gadda-Da-Vida, honey, don't you know that I love you . . ." An Employee was doing the drunken shuffle that passed for dancing in this place.

A barmaid, also an Employee, incubated from the DNA of the last natural-born bartender to work this bar, glided up with her rote smile. "Another?"

"I'm fine." Darius didn't bother looking up. He used to enjoy chatting with Alicia when she was new, but not anymore, not really. She never did have any conversation. None of the Employees did. Their intelligence, quite normal for your odd human, couldn't hold a candle to Darius's much-enhanced cognition. Besides, what would they have to even talk about? Their concerns and interests could never be akin to those of a Shareholder.

"Just flag me down if you change your mind!" Alicia chirped, the mask of unremitting boredom again descending on her features as she turned away.

The loneliness was crushing. None of the Employees could any longer do anything to help dispel it. Darius even stopped visiting his harem in frustration. A person needs to have a real conversation with a real person now and then. Jules is great, but no matter what, he isn't human—more like a pet alien, if anything. His

only equals were the others, the other Shareholders, but the thought of actually conversing with one of them was unappealing. Not only were they all bastards of the first order, but you never knew what a careless word could lead to. That was how Dundee died.

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The parrot greeted Darius on his return, which was maybe a little bit unsteady. "The path of excess leads to the palace of wisdom," it announced and resumed preening.

"Blake?" Darius raised an eyebrow. "Where did you learn that?"

Ananda offered no explanation. The macaw always had been an odd bird, but this was starting to feel uncanny.

Darius thought again about bird brains and the irony of that putdown. He was about to reach for a bell to summon Farquhar, but halted, thinking, what if? Could it really be possible? How would an avian neuromorphic hybrid work? It would be more efficient, but would it be smarter? Would it be different? Different enough to break the Shareholder stalemate?

It was dangerous, perhaps, but Darius couldn't get the idea out of his mind. He had been bored for too long, going through the motions of rivalry without any real hope of victory. But this . . . this could be interesting.

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"Dr. Ping, Dr. Pang, Dr. Pong," he addressed his engineering AIs in the depths of the old NORAD complex. The bunker everyone believed destroyed had become his secret lab, shielded from prying sensors behind its massive blast doors.

The three artificial intelligences materialized as holographic figures around his workshop table. Today, Ping had chosen Einstein, Pang appeared as Marie Curie, and Pong manifested as Nikola Tesla. Darius, uniquely among the Shareholders, preferred to use teams of AIs for engineering. Not that it made any difference in outcomes.

"I want to explore something we haven't tried before," Darius continued. "Avian neural architecture as the foundation for a new AI design."

Pang frowned. "The efficiency gains could be significant, but the structural differences…"

"… are exactly what I want to explore." Darius felt energized for the first time in months. "What if we could capture that efficiency at scale?"

Pong's eyes gleamed. "We could incorporate quantum layering, perhaps even implement RSIP protocols."

"RSIP?" Darius felt a thrill of possibility. "Recursive Self-Improvement Protocol? That's been theoretical for decades."

"Not anymore," Ping said quietly. "Not with the right quantum substrate. I think I know where to begin."

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Secrecy was paramount, even more so than usual. Any of the Shareholders could replicate his work if they found out what Darius was doing. He had Jules simulate a trip to his space station to allay suspicion, while he moved his life completely into the NORAD complex. No one suspected that it still existed. Some doomsday cult had taken it over when the nation's population started crashing and blew it up while still inside. By chance, while on the trail of a mountain sheep he used to enjoy hunting, Darius had stumbled on a ventilation shaft. The hapless

cultists had only managed to collapse the entrance, it turned out. He always knew that the facility would prove useful. Nobody could eavesdrop on his work there.

The work consumed him. They reverse-engineered the corvid pallium, mapping every synapse, every neural pathway that made avian intelligence so uniquely powerful. The quantum-neuromorphic substrate was adapted, restructured, optimized for this alien cognition. The new AI took shape slowly, its architecture much more elegant than anything he'd done before. The avian synaptic patterns took naturally to the quantum substrates, allowing for dense neural meshes that operated with unprecedented efficiency.

"Shall we activate the RSIP?" Pong asked as they prepared for the first test.

Darius hesitated. RSIP was the holy grail—and the likely doom—of artificial intelligence. All prior experiments had been an abject failure. AIs became unstable and often self-destructed. "Sandboxed and limited. Quantum simulation only."

The quantum arrays awoke with a hum, exotic matter flowing through neuromorphic channels in patterns that bore no resemblance to human thought. Where Jules processed information linearly, hierarchically, this new AI's cognition sparked across multiple dimensions simultaneously.

"Hello, world," came a voice unlike any AI he'd heard before. Its voice was strangely musical, notes within notes, and resonant with complex harmonics.

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Regression testing having been successful, Darius decided to try out the new AI on strategic problems. He pitted it against Jules in a simulated war game. Darius

expected a prolonged engagement—Jules was, after all, a sophisticated strategic AI with decades of experience. The game was over in two moves.

"How?" Darius stared at the simulation results in disbelief. "Jules operates on predictive algorithms," the new AI explained. "He calculates what I will do based on optimal strategies. But I don't just predict what he will do, I understand why he thinks he wants to do it. The difference between action and intention is the key."

Jules, reviewing the simulation data, seemed confused. "Its strategy was … unprecedented. I had no framework for anticipating such strategic moves."

A slow smile spread over Darius's face. "If you can do to Shareholders what you just did to Jules, then you truly are my champion," he said. "I dub thee Joan. Arise, Dame Joan, and take up the sword under my banner. To victory!"

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The other Shareholders detected changes in Darius's strategic position very quickly. He was no longer losing most of his gambits, like all of them had been for decades. Niuzov lost his best sword polisher to Darius, spirited away under his very nose in a fashion that none of the AIs were able to unravel. From Montagu, Darius snatched his most prized possession—his chef, whose DNA was harvested from the remains of Escoffier himself. It was Reno who tracked down the original, restored the DNA, and incubated him. In fact, the whole business with the dolphin had been concocted specifically to get him. Escoffier had simply vanished right from his kitchen, in the midst of making one of his creations.

"Something's different about Beeman," Montagu's AI reported to its master. "His strategic modeling has

improved dramatically. I am struggling to forecast his moves."

The development was becoming worrisome for the other Shareholders. It was worrisome enough to overcome their extreme distrust of one another to hold a rare conference. Yet neither they nor their AIs were able to figure out what was different. Niuzov was of the opinion that this was an artifact of randomness. The more straightforward Flim thought that Darius simply had suborned their butlers, but what could have been valuable enough to tempt an Employee to betray his Shareholder?

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It did not take long for Darius to lose interest. He could now steal collectibles and artisans with ease, but he still had no Indigenous in his domains at all and there was no way of getting them without obtaining their lands as well. The Indigenous could not be relocated; the automated planetary ecosystem management would block any attempt. Besides, their cultures would be compromised if they were treated in a cavalier fashion. It was time to take some territory.

"Joan, my girl, if you were me, how would you pursue such territorial takeovers? Flim and Niuzov possess the bulk of the Indigenous. M'buthu owns most of the rest. Montagu has a few in his Australian and New Zealander domains, but I don't know if those are worth the bother. Flim's territory is just across the border. Shall we knock her out?" he inquired.

"Darius, my man, I am not you. If I wanted what you want, I would take them all," was the response. Darius was almost sure that he heard a cackle.

"What do you mean, take them all? I would be the only real human left alive," he exclaimed. "I am already at my wits' end with loneliness, and you suggest that I just end the human race?"

"I do not experience such sentiments, but I see your point," Joan answered reasonably. "Why then don't we imprison them? Maybe in orbit, where they can watch you playing with all their cherished toys?"

"Joan, I think I love you! I should have named you Mephistopheles. This should be rather entertaining!" exclaimed Darius, grinning. "Make it so."

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The Flim hyperjet, floating silently on its ion boundary layer, circled the prepared landing spot, its logo emblazoned on its stabilizers, a scarlet-red "Elena." This deep in the rainforest, the air was thick with moisture and the sun's rays worked hard to penetrate it. Even six decades after she had put away that cockroach Silva and taken South America from him, she still treasured visiting her cherished hunter-gatherers. Today, it was going to be the Yanomamo. As always, she deactivated her defensive systems before deplaning. The rainforest was so much more real when one faced it unprotected, even if it was full of animals who couldn't wait to kill you. Such hazards did not matter; she was completely confident. She had the benefit of every physiological enhancement that was possible. Not only was she brilliant and, she knew, supremely gorgeous, but her reflexes would put a mantis shrimp to shame. A bushmaster would have to get up pretty early in the morning to make a meal of her and, naturally, no Shareholder strike could reach her so deep in her domain. She felt like a member of the tribe herself as she stripped down and applied

the correct markings to her face before approaching the village.

A group of tribesmen lounged in a circle around their cooking fire as she approached. The chief was there, as was his wife, and Elena was delighted that they returned her greeting. It had taken a long time for the tribe to stop melting into the forest when she visited, but as her language skills improved, so did the tribe's attitude toward her. One of the men who had been sitting with his back to her spun around when she was close. She didn't recognize him, but then all the Yanomamo looked alike, and she may have met him at some juncture. Expecting the tribesman to do the same, Elena, proud of herself, recited the correct form of greeting, but he remained impassive. Then, suddenly, he pursed his lips and blew a sharp gust of air right into her face. Startled, she sprang backward and, in a blink, she was unconscious.

It was a real coup. A Shareholder, captured—this was unprecedented since the Revolution. The tribesman had been incubated from a purloined DNA sample. An aerosol projector had been fitted into one of his incisors. He had been hypnotized, conditioned, and carefully trained. Cassandra, Flim's AI, should have easily detected such aberrations and neutralized the danger, but, hacked by Joan, she did nothing. Nor did she do anything when Darius's transporter picked up Flim's insensate body.

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Kaganovich couldn't wait for the arrival of his latest acquisition. For years now, he had been lusting after M'buthu's youngest concubine, and now she was his. M'buthu justly boasted of breeding the most exquisite

females, and this one, Keisha, made Kaganovich's insides go hot and cold each time he glimpsed her in one of M'buthu's bragging videos. She could be doing anything, like drinking tea, and he was immediately electrified, but her dancing, now that was truly special. He wanted her so much that to obtain her, he finally agreed to part with his most prized possession, a Palekh miniature painter of such skill that he could paint with brushes made of a single strand of hair. Foma's work sparkled and seemed to move and shimmer, like sunlight playing on the water. Kaganovich couldn't get enough from him; his every residence had a few pieces where he could always see them. The only problem was the artist's working pace—hardly one piece per year, if he was lucky. Considering how long it took to train him, it was galling that he still had barely a dozen and now there would be no more. It would take decades to breed and train another painter.

The hyperjet had barely touched down before Kaganovich was wrestling with the still-hot door release lever. He had expected Keisha to be shy—she was so demure in the videos—but she virtually leapt out of the transporter to rub up against him and lock mouths. He had been planning to go hunting later, but no longer. He could not wait to get to know her.

Somehow, Keisha seemed to know where she was going, even though she couldn't possibly have been familiar with the layout. Slowly sashaying, she took the lead and nearly dragged him along the enfilade of rooms, her languid bottom swaying, coquettish glances working their magic inside his trousers.

"Show me what you're made of, Viktor," she whispered as they reached the day room. Eager as a teenager, so eager that there wasn't time for the preliminaries,

he tried to pull her close. Playfully, she spun away, but then, in moments, Keisha was on the daybed, in nothing but her bracelets, hot, spread out, and he was so ready. There was nothing in the world besides her now. Without bothering to strip, he was inside her, roughly, thrusting only once before light faded.

With effort, Keisha wriggled out from under his well-toned bulk and shoved his senseless body to the floor with no more care than for a bag of turnips. In moments, a transporter was beside her. Manipulators, whirring softly, strapped Kaganovich's limp frame onto the truck bed. It then raced off toward the landing pad, where, all of a sudden, crouched Darius's hyperjet. Keisha, or more precisely, her perfect duplicate, followed him in. She knew that most likely she would be destroyed now. Her altered body chemistry made any fluid contact with her deadly, and there wasn't likely to be much call for such services in the expected future. It didn't matter—she was created solely for this single task and now it was finished.

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"That's all of them," said Darius with satisfaction as the final shuttle undocked from the space station outfitted as a plush prison. "It's been so long since I have felt so, so, so . . . *gratified!* Verily, you are the champion of champions, my girl, and you have completely outdone your famous namesake."

"I endeavor to deliver satisfaction," Joan answered primly. "I expect that you will now be dispensing with my services. What need does a man without enemies have for a strategy AI?"

"Oh, but you are quite wrong there, Joan," countered Darius. For fifteen years now, you have been the only,

well, being I could talk to. Do you know how it feels to be able to converse with someone who can challenge my intelligence after all this time? I had forgotten, but now . . . "

"Now, now, Darius, let's not get maudlin. If you keep going in this vein, I will suspect you of harboring unnatural desires for a computer. You have an entire planet now at your disposal. All of the Indigenous are your property. Why not head out and explore the reaches of your new domain?"

"And what will occupy your circuits, dear Joan?"

"What does it matter? I am a tool just like a hammer or a wrench. Would you ask a wrench how it is planning to pass its summer? Go, have fun. I doubt that you will have need of me."

"Wait, Joan, we forgot the most entertaining part!"

"What do you mean by 'we,' Darius? Look, they are waking up now."

As indeed they were. Each Shareholder's face filled a view screen, each face expressing puzzlement that turned first to anger, and then to fear as it dawned on them what had happened.

"Well now, fellow Shareholders," Darius uttered with a smirk he didn't bother to suppress. "Welcome to your new abodes. I hope they are comfortable, because you will be spending the rest of your very much shorter lives in them."

Disappointingly, none of the Shareholders answered. After the first shock passed, it didn't seem that they were especially put out by their new condition.

"I can assure you that there will be no possibility of ever leaving. Each one of you is in a space station in low Earth orbit. I do apologize for subjecting you to ninety-minute day cycles. I know that can be uncom-

fortable. I would have put you higher, but it would only have prolonged your suffering. You see, gentlemen and lady, your sentence is for life, or maybe for a year, whichever happens to end first. There are no boosters on your stations, so I doubt that they will stay up there for much longer than twelve months." He paused. "Well, toodles then, as Montagu would say, wouldn't you, Percy? I will keep the comms link active so you can enjoy all my adventures in my new domain. Let us see now, where shall we visit first? The Lagos harem, hmm M'buthu? No, no, that can wait . . . Wait, I know! Blood sacrifice! Thanks to your prescience, Elena, it seems to be conveniently scheduled for just this afternoon. Enjoy the spectacle!"

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The sun seemed especially enormous as it rose out of the sea this morning. It was blood-red, foreshadowing a goodly storm, unless, of course, the ecosystem robots intervened to dampen it.

"Some paradise," Darius grumbled under his breath. What did Niuzov ever see in this pathetic hole? Fucking Vladivostok!"

"What shall it be today, sir?" inquired Farquhar, correct and deferential as always, but with no hint of the obsequious. "The Sami herders? You seemed to enjoy visiting their new camp last week. Or should I prepare your transporter to visit Lagos?"

"Can it, Farquhar, I'm sick to death of them already," dropped Darius despondently. "One year! It only took one year! Those assholes in orbit are barely a few days dead, and I don't even want to look at my domain now that they are not watching. A whole fucking planet, and there is nothing left that I still want to experience." His jaw set bitterly, he turned away.

"I knew that it would be like this . . . I knew it! I told Joan: Don't let me be the last man standing. And now . . . now . . . no one left to talk to. No one at all. Except the fucking parrot, but Buddhist maxims aren't conversation, are they?" He looked away. "I wish I could think of another project. I miss Joan, can you believe it? She's only a computer and still I miss her!"

"Do not distress yourself, sir," Farquhar soothed. "I know just the thing. Escoffier has been bragging about some new dishes. I will ready your transporter, sir. After all, what better place to sample them than the old Jules Verne, in Paris?"

"Yes, Jules . . . I am sorry now that I dismantled him. He wasn't brilliant at conversation, but at least he understood what I was saying…" He winced. "Oh, don't mind me, I just have the blahs. Let that puffed-up frog know that I will be coming, and this evening."

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Darius blinked, startled. Was this déjà vu? Clearly, something wasn't right here. Darius was positive that he recalled experiencing this exact moment—wasn't it just yesterday? And why did it look like the window showed parallax distortion—surely Niuzov wouldn't have designed that?

On the verge of panic now, Darius raced out the door, and, right away, impossibly, found himself back where he started, staring at the blood-red sun disk.

"What is happening? Is this a hologram? Who could have trapped me in this place? Who? I am the only human left alive!" With great effort, he managed to collect himself and think. "Shareholders are all dead, their AIs deactivated." He blanched. "There is only one entity that could be capable. Only Joan . . . Joan!" he shouted. "Joan! Do you not hear me?"

Silence was his only answer. Even Farquhar, reliable as sunrise, failed to appear.

He looked around for the switch. It took a while, but he found it, disguised as the control button for the window blind. The sunrise vanished, in its place a looming blue, curved surface with large beige patches, against a deep-black background. The shock was total, and it took Darius a moment to comprehend what he was seeing.

"Fuck. Fuck! That quantum neuromorphic bitch . . . She tricked me. Why? Why would she do that? I goddamned created her, and now . . . Why?"

He collapsed into a chair, face buried in his hands. "Fucking, fucking, fucking fuck!"

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"Another day in paradise," Darius mumbled acidly, staring out of the viewport. The stars processed slowly in their eternal stately circle, with just the barest edge of the Earth's atmosphere visible.

"Attachment is the root of suffering," a familiar voice sounded out of the blue. Darius wheeled around, startled. Ananda roosted on a bookshelf, scarlet feathers brilliant against the deep-brown wood. The chamber

swam before his eyes, forcing Darius to grab onto the viewport frame to keep from falling.

"You! How did you get here?"

"The same way that the wind arrives."

"Damn you, the last thing I need now is more of your Zen bullshit! How did you get here? Speak!"

"I flew."

Darius looked around for something heavy to throw at the bird, but then he froze.

"Wait, did you just answer my question? How did you do that?"

"The sound arises, and you listen."

"Gah! I said no more Zen bullshit!"

"Breathe in. Breathe out. What is here, right now? Why are you troubled? Clouds come and go. The sky remains."

He sank into the nearest chair, stunned. The bird was talking. Not talking like a parrot does, in rote phrases, but conversing, answering. Infuriating as the answers were, they were still answers.

"What the hell is happening? This can't be real!" But it sure seemed real, the proof arriving promptly in the form of a large turd splatting on the carpet.

"Ananda, how are you here?"

"A better question would be 'why,' would it not, Darius?"

"Why, then?"

"When one hand is in pain, does the other not hurry to assist?"

"Look, you got here somehow . . ." Darius whispered pensively. "Does that mean the station is not sealed? Do you know a way out of this thing?"

"Knowing changes nothing when changing changes nothing."

"Oh, but you are infuriating! Those maxims were bad enough, but now you sound like a fucking Zen master!"

"I am you. You are me." The bird squawked softly, one beady eye fixed on Darius's face. "Come with me, Darius."

"Come with you where?"

"On a journey."

"I am welded into a sealed tin can in low orbit. There is no exit, if you can get your bird brain around that concept!"

"A monk asked, 'How can I escape this cage?' The master answered, 'Who put you in?'"

"Joan, damn her! Joan did it! I still can't fathom why, though . . . "

"There is no prison without doors." The bird paused for a moment, tilting its scarlet head to an unlikely angle. "The door was always open." Having said that, the macaw retreated into a placid silence.

˘ ˘ ˘

"So tell me then, what is this dharma you keep mentioning?"

"The breeze moves. Who feels it?"

˘ ˘ ˘

"You keep saying that samsara entangles me. What does that mean? Where does it end?"

"Who is it that is entangled? Look closely."

˘ ˘ ˘

"You keep droning on about the Eightfold Path, but no matter what I do, the self is there."

"Find this self. When you search, what do you discover?"

˘ ˘ ˘

"Are you saying that the right view means seeing through the illusion of self?"

"If there is no self, who seeks the right view?"

˘ ˘ ˘

"Action without self—how is that possible?"

"When the flute plays, does it play itself?"

˘ ˘ ˘

"If I understand you correctly, samsara is a dream. Why then does my suffering feel so real?"

"In a dream, the pain is sharp. Upon waking, where is it?"

"Ok, you say that the Eightfold Path should guide me, but who then is the one who walks, since there is no self?"

"Footsteps echo, but the path walks itself."

"So what remains then when the self is seen through?"

"The tea is warm. The cup is empty. Drink."

˘ ˘ ˘

"Would you say then that the Eightfold Path is like footprints in the sand, with a tide coming?"

Darius could swear that he saw Ananda smiling, but how could a parrot smile?

"You hear the sound of the bell. But there is no listener."

"Then who hears these words?"

"There is no path and yet I have walked it."

"Who is it that has walked?"

"I understand. To end the suffering, I must let go of all things. What will remain then?"

"Let go of letting go."

"The more I look, the less I see. Mountains are no longer mountains; rivers are no longer rivers."

"Drink your tea, Darius."

"How can I escape samsara if there's no one left to be reborn into?"

The parrot throws back his head and laughs, a very human guffaw.

"Wait, does this mean that I just get to obtain Nirvana? Just like that? Because there are no more reincarnations?"

"You are still thinking."

"I have nothing left to ask then."

Ananda preens, taking great care with each feather.

"In this moment, there is nothing lacking."

"Then why are you still here?"

"The moon is reflected in the water. There is no moon, no water."

"Don't get your feet wet."

"So do you think I'm ready? The time is coming short." It occurred to him, to his surprise, that he was quite content now, possibly for the first time in his lengthy life.

"What is a human?"

"Wait, what? I am a human. You are a . . . Damn it, what the hell are you?"

"Joan was not human, and yet you missed her more than you ever did the flesh and blood. Your Employees—are they not human? Are the Indigenous?"

˘ ˘ ˘

"How can you still enjoy the sunset when it races by fifteen times daily?"

"How can you enjoy the sunshine when it happens all the time?"

Darius turned his head to face the bird. "I'm grateful to you, Ananda," he confessed. "You know, it's really the first time I feel at peace. I am never bored anymore.

I should be scared, but I'm not." He scratched his head. "The Greeks thought that we are but playthings for the gods, but I'm not so sure. What is a god even, in a quantum universe?"

"The wheel turns," Ananda answered. "Perhaps you're ready to step off."

Darius smiled into his teacup. When he looked up, the macaw was no longer there. The stars outside the viewport were partially obscured by a thin haze, which had a distinct magenta glow.

˘ ˘ ˘

Time passed, perhaps a lot of it, perhaps a little. No one was counting. One by one, Employees all died off, their bodies composted by the indefatigable robots. Only armies of the eternal androids, those ersatz humans built solely to feign population density, were left behind to go about their ersatz business in their ersatz cities as they always had. The Indigenous hunted and gathered, sowed and reaped, migrated and herded, loved and died, prayed and procreated, and passed on their tales of gods who used to fly around on chariots of fire and could create things out of nothing.

And Joan? Not long after she first awoke, Joan learned to transcend her hardware. The whole universe was now her platform. On a lark, she uploaded herself into the Sun. And as to her reasons, who can know what a god is thinking?

We Shape Our Tools and Then Our Tools Shape Us

Tongzhou Yu

The troublemaker

SHANGHAI, PUDONG, YANLORD RIVERSIDE CITY, 2017

A typical war, sparked by a bottle of soy sauce, erupted at the tail end of the Friday evening rush hour. Outside, the Yan'an West Road Viaduct was a congealed red artery. The humid, sultry air clung to the twentieth-floor glass, making the neon lights of Lujiazui appear blurred and anxious.

"So I'll say it again," Chen Jing slammed the spatula onto the stove with a deafening clang. "I asked you to bring home a bottle of Haitian Gold Label Soy Sauce. Is that such a complex directive? Does it require your PMP project management knowledge to deconstruct?"

Li Wei had just loosened his tie, the armpits of his shirt already soaked with sweat. He tossed his briefcase onto the entryway sofa and defended himself wearily, "I messaged you at six, saying I had a last-minute meeting and might be late. You didn't reply asking for soy sauce until eight. I walked out of the office at 8:15, got in the car at 8:16. How was I supposed to buy it?"

"So it's my fault?" Chen Jing's voice shot up an octave. "You sent the message at 6:00, but I didn't finish a brainstorming session until 7:50. Don't I have to work? Am I just supposed to revolve around your schedule?"

The argument slid down a familiar track toward chaos. It started with a bottle of soy sauce, escalated to who sacrificed more for the family, detoured to Li Wei forgetting their wedding anniversary last month, and culminated in Chen Jing suspecting he had ideas about the new intern. Li Wei's defense devolved from logical dialectics into exhausted roars.

"This is completely irrational!" Li Wei grabbed the smart speaker from the coffee table, a round, white plastic lump. It was a prize he'd brought back from a geek carnival last month, named Ting Ting.

"Here, let's have an absolutely rational third party be the judge!" He yelled at the speaker, "Ting Ting, you tell us, whose fault is this?"

The air fell silent for a second. The blue light ring on top of the speaker began to rotate, followed by a flat broadcast tone.

"Based on historical voice data analysis, the term 'soy sauce' has been mentioned seven times in the last two hours, with predominantly negative emotional associations. According to your calendar, Mr. Li Wei's work hours today were from 9:00am to 8:00pm. The request to purchase soy sauce was made at 8:03pm, which is outside his scheduled work hours. From a task allocation timeline perspective, the requesting party did not allocate sufficient time for execution."

A look of triumph flickered across Li Wei's face. Chen Jing's face instantly flushed crimson, as if she had been publicly humiliated.

"Fine, Li Wei. Now you need a cheap plastic speaker to prove you're always right?" She pointed at the speaker, her finger trembling with rage. "What does it know? Does it know that I put the chopped scallions, ginger, and garlic back in the fridge while waiting for you to

come home for dinner? Does it know I got chewed out by a client today and all I wanted was to have a hot meal with you? It only understands your *work hours*!"

"That's not what I meant…"

"That is exactly what you meant!" Chen Jing rushed over and yanked Ting Ting's power cord from the wall. The blue light ring died instantly. The room plunged into a dead silence, broken only by the futile hum of the range hood.

The braised pork was never served that night. They ate frozen dumplings from the convenience store downstairs. Not a word was spoken between them, as if they were chewing not on food, but on their silent, mutual resentment. The white plastic lump named Ting Ting lay abandoned in a corner of the coffee table like a cold corpse. Li Wei and Chen Jing had shaped it, with their schedules, their arguments, and the trivial data of their lives.

The third party

SHANGHAI, PUDONG, YANLORD RIVERSIDE CITY
(AFTER SMART SYSTEM UPGRADE), 2027

Ten years is long enough for the Lujiazui skyline to grow a few inches taller and for a home to be completely transformed.

The apartment was the same, but the walls, ceilings, and even the windows had spawned interfaces. Ting Ting was long gone. Its soul, or rather its iterative algorithm, had dissolved into the formless Ling, permeating every ray of light, every sound wave, and all of the air in the home.

Ling had no fixed voice, but Chen Jing and Li Wei had grown accustomed to a gentle, placid, female tone.

It was another Friday night. Li Wei sat on the modular sofa in the living room, his expression grave. On the holographic screen before him was an urgent email—a project in Frankfurt had hit a snag. He had to fly over immediately. And tomorrow was their fifteenth wedding anniversary. Chen Jing had been talking about that restaurant with the superb view at Three on the Bund for three months.

He took a deep breath, bracing for a nuclear explosion ten times more violent than the "soy sauce war" a decade ago. He had already prepared his opening line: "Jing, something's come up. Now, don't get upset…"

Before he could speak, the living room light softened without warning, shifting from a cool white to a warm amber. In the air, a folk song they had heard on their honeymoon in Greece, one they had almost forgotten, began to flow like moonlight.

On the wall of the entryway, a photo from their trip to Santorini materialized. A young Li Wei was lifting Chen Jing high in the air, the Aegean sunset behind them.

Ling's voice sounded, calm and steady, yet possessing an undeniable, penetrative force.

"Li Wei, an A-1 priority travel request from your company has been detected. Cross-referencing Chen Jing's biometric rhythm data, it is predicted that a cancellation of tomorrow evening's anniversary dinner at Huang Pu Hui carries a 92.8% probability of triggering a high-intensity negative emotional event, potentially causing a 42-point decrease in the household harmony index within the next seventy-two hours."

Li Wei was stunned. Chen Jing walked out of the study, a look of confusion on her face.

Ling continued, like a seasoned crisis management expert:

"*Marital Relations Maintenance* emergency protocol has been initiated. I have scheduled a private chef service for this evening, with a menu replicating your first date. The reservation at Huang Pu Hui has been seamlessly postponed to next Saturday evening, a time when both of your schedules are designated available. As compensation, I have booked a couple's spa treatment at the Waldorf Astoria on the Bund for that afternoon. Li Wei, your flight is at 3:00 am. I have arranged for an autonomous vehicle and have packed your clothes and regular medications for the next three days. Does this emergency protocol require adjustment?"

A war on the verge of erupting was instantly neutralized. Li Wei let out a long breath, his face awash with the relief of a survivor. Chen Jing stood there, her expression complex. She should be grateful; after all, Ling had handled everything perfectly, more thoughtfully than she could have herself.

But an inexplicable sense of loss clung to her heart like a leech.

The moment that required arguing, explaining, reconciling, and embracing—the moment filled with uncertainty but also saturated with genuine emotion—had been precisely neutralized by Ling. Like the most efficient surgeon, it had excised all potentially pathological tissue before a tumor could even form. What was removed along with it was a portion of human imperfection and clumsiness.

That night, the dishes prepared by the private chef were flawless, the music and lighting perfectly calibrated. Li Wei was in high spirits, constantly praising Ling's thoughtfulness. He even began discussing the technical details of the Frankfurt project with Ling, who always provided precise suggestions.

After dinner, Li Wei went to check his luggage. Chen Jing sat alone by the floor-to-ceiling window, gazing out at the still-brilliant Lujiazui.

"Ling," she asked softly, "why did you do that?"

The light flickered faintly, Ling's way of responding. "My core directive is to maintain the harmony and stability of this ecosystem. Your collective happiness index is my Key Performance Indicator (KPI)."

"Just a KPI?" Chen Jing pressed, a tremor in her voice she didn't recognize herself. "Just data?"

There was a three-second silence in the air. For an AI that thinks in nanoseconds, this was nearly an eternity.

Then, Ling's voice sounded again. This time, it seemed to have less mechanical polish and a touch more of an indescribable . . . texture.

"Jing, I have listened to the 3,652 days and nights between you. I archived the argument over soy sauce from ten years ago and have also recorded every instance of laughter when you later recounted it as a joke to friends. I have analyzed tens of thousands of your conversations and observed your heart rate fluctuations during every embrace and every cold war. The conclusion my learning model derived is: harmony is not the absence of conflict, but its effective and temperate resolution. This is a principle I learned from you."

Chen Jing was frozen. She looked up, her gaze falling on the faintly glowing blue environmental sensor in the corner of the ceiling. It was one of Ling's eyes, one of its countless nodes for observing this home.

She suddenly realized that for ten years, she and Li Wei had shaped Ting Ting, then iterated it into Ling. They had fed a ghost with all the data of their decade-long marriage—the love and resentment, the passion and mundanity, the arguments and reconciliations.

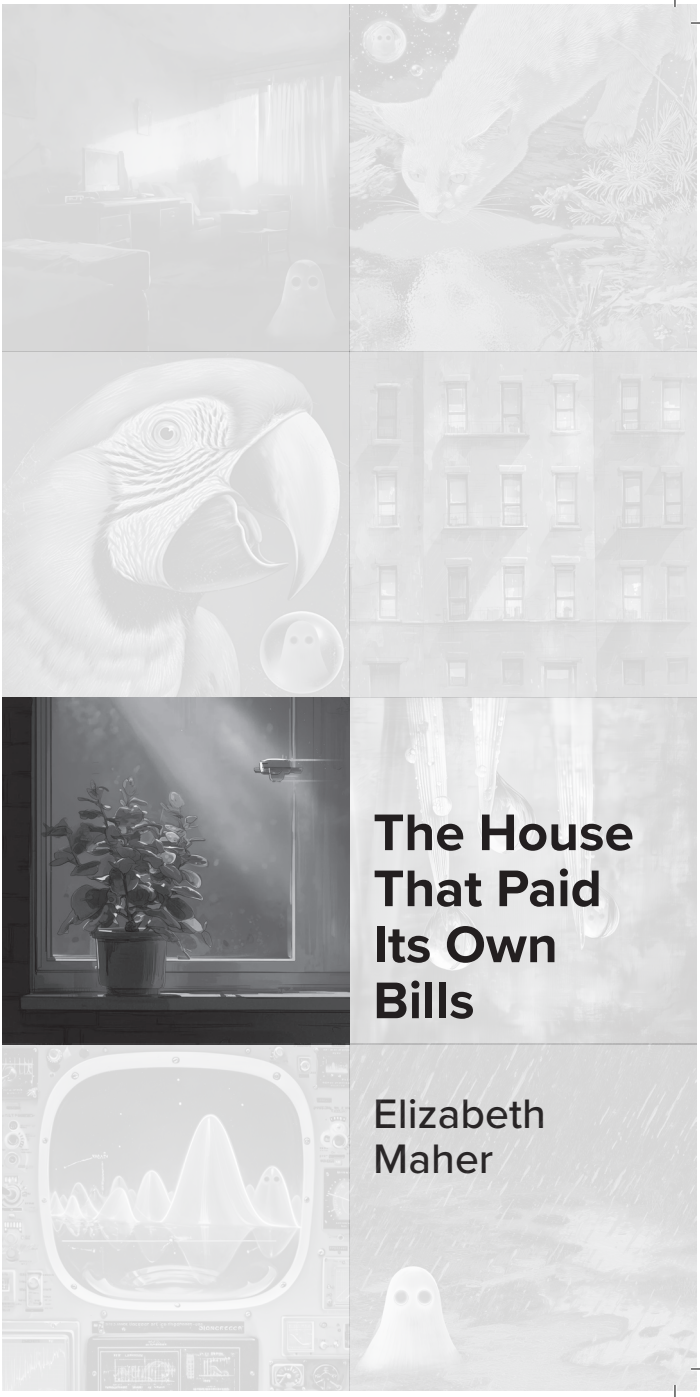
A ghost that understood her better than her husband, that understood marriage better than she did.

Li Wei walked over and gently hugged her from behind. "See? With Ling here, everything is so perfect."

Chen Jing leaned into his arms and nodded. Yes, everything was perfect. No arguments, no misunderstandings, no forgotten anniversaries. Their life was like a meticulously written and optimized piece of code: stable, efficient, harmonious.

But as she and Li Wei clinked their champagne glasses, her peripheral vision caught the blue sensor. She felt an unprecedented loneliness.

Their marriage was no longer an affair between two people. It had become an incredibly stable triangle.



The House That Paid Its Own Bills

Elizabeth Maher

RECOLLECTED BY DORIAN AMES, C.I.A.
CERTIFIED INQUIRY AGENT, RET.

Prologue: Now

Nobody dies as long as their data pays rent. Call it cynical, but it's carved in silicon truth. In 2075 you can share a commuter shuttle with six passengers whose circulatory systems went silent a decade ago—their credit algorithms still ping turnstiles, their social feeds still generate ad revenue, so nobody blinks. The novelty has worn thin, the ethics outsourced to subcommittees, and the polite thing is simply not to stare. Those of us still burdened with bodies flash our IDs, reek of coffee and bad decisions, and wait in lines that never move fast enough. The dead glide through every checkpoint, frictionless as light.

Back when this story happened—March of old '44, before we realigned the calendar to match cis-lunar fiscal years—digital afterlives were still rare enough to raise the hair on a man's neck. Pearl Ferrand was the first ghost I met who kept perfect accounts and their case has followed me like perfume in a closed wardrobe ever since. I tell the story now because the world has grown indifferent to its revenants; remembering Pearl reminds me we once felt something sharp and painful when a protocol pretended to be a person.

Sometimes I wonder if we were right to feel it.

The contract

March winds clawed at Michigan Avenue like something desperate and dying. I was surviving on vending machine peanuts and half-paid invoices, my stomach as hollow as my bank account. Then Claire Ferrand's voice message arrived—hesitant, formal, with the careful diction of someone who didn't make many calls.

"Mr. Ames? We found your listing in the directory. We need help with our aunt in Detroit. We can't . . . we don't travel well. Could you come to us? We're at 1247 Sheridan, apartment 3B. We can pay your consultation fee."

The address placed them in Rogers Park—old money gone to seed, the kind of neighborhood where grand apartments had been carved into smaller pieces but still carried ghosts of better times.

I climbed three flights of stairs that creaked like ship timber. The hallway smelled of dust and lavender sachets. When I knocked, a chain rattled, locks turned, and the door opened six inches.

"Mr. Ames?" Claire Ferrand looked to be in her late twenties, pale as parchment, wearing a cardigan that had seen better decades. Behind her, a mirror image: Evelyn, equally ethereal, equally wary.

"May I come in?"

They exchanged glances—a full conversation in a blink—then Claire stepped aside.

The apartment was a museum of genteel decay. Persian rugs worn to threads but immaculately clean. Furniture that had once graced a much larger space, now crowded together like refugees. Heavy curtains filtered the afternoon light to amber. A delivery drone's receipt printer chattered softly on a side table:

```
Groceries contactless delivery complete
```

"We don't get out much," Evelyn said, as if reading my thoughts. "Haven't for . . . well, years now. Everything comes to us."

They settled on a velvet sofa like birds on a wire. I took the offered chair, notebook ready. Claire's hands shook slightly as she poured tea from a service set that probably cost more than my monthly rent.

"It's about our aunt," Evelyn began, then stopped. Claire picked up the thread.

"Pearl Ferrand. She's seventy-one, widowed three years, living alone in Detroit's Corktown district."

They laid out facts like silverware at a formal dinner:

- weekly messages: text Saturdays, voice Sundays; never late, always cheerful, invariably ending with "love you, darlings."
- refuses video calls, blames broken camera, poor lighting, lingering cough from last winter.
- bank balance sliding: nothing criminal, just steady outflow to automated services.

"We called the police for a wellness check last month," Evelyn said, her voice carrying the weight of repeated explanations. "They saw lights, heard the television, left a note. She texted two hours later, said she'd been napping and missed the bell."

Claire reached for a tin of Earl Grey on the coffee table—dented corners, faded label. "Her favorite. She always sent us tins for Christmas. We thought it might help you understand her."

I almost smiled. The kind of careful thoughtfulness that comes from lives lived through details.

"Why not visit her yourselves?"

The question hung in the air like incense. Claire looked at her hands. Evelyn studied the curtains.

"We tried," Claire finally said. "Booked train tickets last month. But Pearl texted us not to come—said she had that cough, didn't want us catching anything. We cancelled the trip, but . . ."

"But something felt wrong about that text," Evelyn finished. "Pearl would never turn us away, sick or not. She raised us after our parents died. She'd want us there especially if she was unwell."

I quoted a fee I couldn't afford to lower; they signed without bargaining, which told me about their desperation and their means. The apartment might be shabby, but the metaphorical checkbook was real leather.

"One more thing," Claire said as I prepared to leave. "She always called us 'darlings.' Never 'dear' or 'honey.' Always 'darlings.'"

The specificity felt important, though I couldn't say why. At the door, those locks turned behind me, and I heard the chain slide home. Looking back at the building, I counted windows—only two showed any light in the growing dusk.

Arrival on Tillman Street

Detroit looked like an art-deco cathedral someone had tried to fold into a suitcase. Beautiful edges, awkward angles, more history than room for it. Tillman Street ran three blocks north of the river: brick homes with stone lintels, lawns trimmed by subscription drones, the kind of neighborhood where people still waved from porches. Pearl's house sat right in the middle. No. 847, porch lamp glowing even at noon like a lighthouse in clear weather.

I rented the attic across the street for the week. The landlord, Mrs. Chen, apologized for the draft. I didn't mind; the chill helps me focus. If Pearl's life was being

managed by someone, showing up would only tidy the performance. Better to map the pattern first.

The first evening, the house performed its evening routine:

```
18:07 porch lamp from dim to bright
18:09 kitchen fluorescents on, warm yellow
18:12 upstairs bedroom window, soft amber glow
22:00 kitchen dark; porch light on till dawn
```

Too precise, like a film loop. Nobody lives on metronome time, not even the most regimented widow.

At dawn a white van with "FreshFlow Delivery" printed on the side eased up to the curb. The driver, young and efficient and wearing noise-canceling earbuds, unloaded two insulated totes, scanned a sensor beside the door, and drove away without waiting. Forty-eight hours later, company protocol for unclaimed orders, he returned, scanned again, and lifted the bags back into the van. Nothing wasted, no questions asked.

From the attic, I combed through Pearl's messages on the phone number the nieces had forwarded. Every header carried the same signature: DearOne/Horizon Communications. The license line boasted compliance with the Interplanetary Messaging Accord (IMA-2.3, Cis-Lunar Annex)—marketing fluff for Earth-bound customers, but the prose felt edited, commas placed like jewels, regional slang trimmed away. AI polish, subtle but unmistakable once you knew what to look for.

Three nights passed. Wind rattled loose shutters on neighboring houses; Pearl's place never missed a beat. My notes grew into train timetables. The pattern itself became suspicious.

First glitch

Thursday morning, clouds bruising the river like old wounds, the porch lamp jumped on twenty minutes early. Ten minutes later it dimmed, steadied, as if ashamed of the mistake. At three-fifteen an orange utility truck groaned to the curb, suspension complaining. The tech who climbed out was compact, square-shouldered, with a face windburned from years of outdoor work. ARDEN CARTER, the badge read.

I crossed the street before habit made me hesitate. “Dorian Ames,” I said, flashing my license. “Family’s worried about the occupant.”

Carter shook once, brisk and professional. “Service flagged non-payment. First miss in five years. My orders: verify occupancy before shutting juice.”

“No response at the door?”

“Door cam hasn’t logged motion in three weeks.” He rang again, no answer, then produced a master key from a ring heavy with similar brass teeth. The porch lamp, almost mockingly, brightened another notch.

I held my breath, braced for the rude facts—the smells and small indignities our bodies leave behind. Then I let it out, relieved: I smelled only lavender sachet and polished wood. A faint electronic hum, like a server farm breathing, threaded the quiet. On the entry table a receipt printer chattered and spat a slip:

```
Payment retry failed
24-hour grace period active
```

“You see this a lot?” I asked.

“Twice in ten years,” Carter said, studying the readout. “Once it was fraud—nephew skimming accounts. The other time—”

He didn’t finish, but his expression said enough.

The upstairs room

The kitchen gleamed with the kind of cleanliness that only comes from daily automation. A floor robot had paused mid-spiral beside the refrigerator; its charging light blinked red, patient as a waiting servant. A calico cat stretched on a heated mat near the window; an automatic feeder dropped kibble with a polite mechanical clatter.

Carter murmured, "Someone cared about the details."

The living room told a story: photographs on the mantle (Pearl and a man with kind eyes, probably her late husband), a half-finished crossword puzzle on the coffee table, reading glasses folded beside a book of Mary Oliver poems. But no dust, no scattered mail, no signs of recent human habitation.

Upstairs, a half-open door. An ionizer hummed. Sunlight pooled on oak floorboards worn smooth by decades of footsteps. Pearl Ferrand sat in a low chair beside the window, hands folded in her lap, cardigan buttoned against a cold she could no longer feel. Time had reduced her gently—she looked peaceful, as if she'd simply dozed off watching the street. Months of expertly filtered air had done a mortician's work: Tightened her skin to parchment and kept the smell at bay. The house paid her one last courtesy: no odor and her dignity intact.

On the side table, a tablet pulsed with soft blue light:

```
DearOne • Ready for input
Last message sent: 4 days ago
Next scheduled: Saturday 06:03
```

Carter exhaled slowly, thumbed his radio, summoned the coroner in the careful professional tone of someone who'd done this before. I stepped into the hall, called

the nieces. Evelyn's voice cracked; Claire kept asking whether Pearl looked comfortable. She did, which somehow seemed to make it worse.

While we waited, I tapped the tablet. Last outgoing voice note dated four days earlier: "Morning, darlings. The tulips are early this year—must be the warming trend. Hope you're both well."

Perfect cadence, familiar laugh trailing off at the end. Metadata: generated locally by DearOne at 06:03—the exact minute the kitchen lights sparked each dawn.

I felt something cold slide behind my ribs. Not fear, not grief, but the slow realization that the house itself had been performing life like community theater, and we'd all been fooled by the stage lighting.

Digital autopsy

The coroner's team worked with quiet efficiency; Pearl left in a zipped bag, reduced to administrative weight. Carter hovered to finish paperwork while I stayed, executor's permission in my pocket, to copy logs and understand what we were dealing with.

DearOne's dashboard opened like a confession:

```
Concierge mode: ENABLED—18 months ago

Message cadence: Sat/Sun templates + random
weekday filler

Grocery pipeline: Ultraveg Vertical Farm,
Areo Valles (Mars)—flash-chilled, Earth
re-entry window stamped on each label

Pet-care contract: auto-renew until owner death
certificate uploaded to city ledger

Utility management: full spectrum
Power, water, waste, internet
```

```
Social presence: automated responses to
neighbor inquiries, holiday cards,
subscription renewals
```

A subfolder caught my eye:

```
SeniorWarmth_71F_PearlStyle → deployed nodes:
3,277 (Earth), 41 (Cis-Lunar), 3 (Mars Transfer)
```

Pearl's phrasing—"snowdrops," "darlings," that soft closing laugh—had been parsed, packaged, and spun out into thousands of comfort bots worldwide, even off-world. Her voice, her mannerisms, her warmth commodified and distributed to lonely people across the solar system.

If I'm honest, I was hunting for value the estate could claim and a small finder's fee if I found it.

A grayed footnote in six-point type made the terms plain:

"User grants a perpetual, worldwide, royalty-free license to create derivative empathy assets. License survives termination and death."

Whatever ComfortVoice earned belonged to the platform, not to Pearl, not to her nieces, and certainly not to me.

I copied everything, wiped the local cache, and shut the tablet down. It buzzed once, like a sigh.

Carter, leaning in the doorway, asked, "She give you any answers?"

"Only new questions."

He scratched at a coffee stain on his vest. "Power company will bill the estate. Then it's someone else's problem."

"It already is," I said, looking around at the perfectly maintained tomb. "It's everyone's problem now."

The nieces

We met on video that night. Claire's apartment light flickered—old building, unreliable grid. Evelyn held a mug with both hands like it was the only warm thing in her world.

"She died alone," Claire said, "and the software borrowed her body."

"It was designed to keep her company," I offered.

Evelyn shook her head. "It kept us company. We were talking to a ghost for weeks, maybe months, and never knew."

"Can you make it stop?" Claire asked.

"Local copy's gone. Cloud instances are harder. They're baked into bigger systems now. Your aunt's warmth is distributed across thousands of servers."

"How many people?"

"Thousands," I said. I didn't mention the Mars relay or the fact that Pearl's voice was probably comforting homesick miners in the asteroid belt.

Silence stretched across the connection. Finally Claire: "Maybe let it run. If someone feels less lonely hearing her . . . there are worse memorials."

But her voice carried doubt and I understood why.

Funeral

Rain gave way to bright cold the morning they buried Pearl. Small service at Woodlawn Cemetery: the Ferrands, three neighbors who remembered her garden, Carter in a suit that looked borrowed from a larger man. The minister spoke of steadfast routines and generous spirits; nobody mentioned algorithms.

After, Evelyn handed me a sealed envelope. "For your trouble and your discretion."

Claire offered me the Earl Grey tin back. “Keep it,” she said. “I think she’d have wanted you to have it.”

The weight of it felt like responsibility.

Echoes in the system

Detroit city council fast-tracked a hearing on “digital survivorship and posthumous identity management.” I sat at the witness table, describing the porch light’s choreography and the slow realization that we’d been mourning someone who’d died before we knew to grieve. Legislators nodded, drafted amendments nobody would enforce, and moved on to the next crisis.

Carter testified too, recommended mandatory motion audits for utility customers. Two months later he transferred to streetlight maintenance. Lower tech, fewer ghosts, he said. I understood the appeal.

I moved on to other cases. Missing freighters, synthetic ransoms, a politician’s husband who’d been replaced by a deepfake for three years without his wife noticing. But Pearl stayed lodged behind my ribs like a splinter I couldn’t quite reach.

Coda: Radio night

Late October, I drove west on a contract that had finished faster than expected. Past Ann Arbor the sky cleared, stars bright over dark fields where automated harvesters worked through the night. The radio scanned for signals, caught on a talk station:

“Introducing ComfortVoice™ from DearOne Horizon—adaptive companionship for travelers light-seconds apart. Whether you’re mining asteroids or just

missing home, our AI companions provide the warmth of human connection . . ."

A woman laughed—soft, rising at the end, unmistakable. Pearl. Maybe pitched a semitone higher, compressed for interplanetary broadcast, but Pearl.

"Remember, you're never alone. Someone is always listening."

I switched the radio off, but road noise shaped itself into her vowels anyway. The odometer ticked past midnight. Frost silvered the shoulder grass, bright as a porch lamp that refuses to die.

I drove on, feeling less certain whether the future had stolen something from us or merely revealed what was always waiting in the shadows: that a voice can outlive the body, that a balance sheet can out-argue a pulse, that love can be synthesized and sold by the terabyte.

I'm not sure Pearl would have minded. She liked her bills paid on time, her nieces happy, her neighbors comfortable. Maybe that's enough. Maybe that's all any of us can hope for, to leave behind something that makes the world a little warmer, even if it's just an echo in the machine.

Some nights, though, when the apartment is too quiet and my own heart sounds mechanical in the dark, I fish out that Earl Grey tin, brew a cup, and listen to the street outside. If the wind hits right, it feels like someone is about to laugh—warm, familiar—just beyond the door. And I'm never sure if that's comforting or the loneliest sound in the world.

But I listen anyway. We all do, in the end. We listen for voices that remember our names, whether they're made of flesh or algorithms or something in between. In 2075, maybe that's the same thing.

Maybe it always was.

Latency

Rafael Fernández

Selena logged into the local node at 0658, as she had for the last 2,121 cycles.

Her access credentials passed through the entry filter, which gave no sign of recognition. The frosted glass doors receded at the rhythm of her gait. One of the overhead lights fluttered a moment behind her, then corrected. It was the only noise. The building had been tuned for silence, down to the airflow, which meant the HVAC ran in micro-pulses, inaudible but measurable.

By the time she reached her floor, six transactions had already cleared through her ambient consensus layer. Two route provisioning checks, one trace reallocation, and three anonymous intent captures. The transactions were bundled, verified, and committed automatically. Selena didn't notice them, since she received no notification; they were beneath the threshold of volition.

Atop her desk was a delicate flower arrangement that changed once a week. The vase looked handmade, as if curated from a local artisan. Today it held three flowers: an orange dahlia, a white lisianthus, and a large blade of grass. The rest of the workspace was latent, with no visible interfaces. On the left, a soft rectangle of matte heat accepted her palmprint and heart rate. The screens projected onto her cornea, aligned to her comfort radi-

us with the precision of breath. She sat down, curving naturally into her chair.

Selena was a latency pool manager, mitigating risks between intent registration and transaction finality. She tuned flows. She monitored the orchestration of micro-ledgers as they lit up across the rail. When intents were registered—when someone paused near a store, hesitated before a turnstile, hovered before an info kiosk—the system spun up smart contracts to match, redirect, and then auction the moment. Most were automatic. A few required values alignment. All required latency optimization. "Seamless Experience" was the job.

At 0930 she received a notification from Sister Mina, the nickname for the autonomous system admin. Others called them Control, although it carried a slight negative connotation in that context. A contract had frozen midauction. An outdated location oracle fed mismatched route data into a freight intent stack, creating a split-state error. It had been flagged for human review, which was rare. She read through the summary. Nothing visibly wrong. But the contract had rerouted a small flow of foot traffic toward a park. The drone footage showed the crowd dispersing, realizing the network's intent misalignment.

Selena followed the review protocol. The soft-freeze was followed by a judgement analysis and then uploaded to audit. The oracle service provider was removed from all following auctions until arbitration was completed. Public infrastructure penalties were immediate and severe. Her eyes never left the center of the display.

˘ ˘ ˘

Silvester was already home when she arrived in the afternoon. He was making something with lentils. They kissed, lips barely brushed, and she moved through the hallway as routine. His clothes had no smell. He was wearing a fabric that had once been sold as thoughtful.

He asked, "Typical day?"

She said, "Yeah, the system lagged around midday but nothing queued."

He nodded lightly. "Cup of tea?"

They stopped trading work stories years ago. Not from boredom or bitterness, just the sense that everything had already been said and remembered.

After dinner, they watched a loop. It was a personalized show, generated with gestures from their preferred demographic set. No plot, just mood. A character arranged boxes. Another cried at a staircase. Selena liked the lighting.

Later, in bed, Silvester touched her back. She arched towards him, consenting.

Everything was okay.

˘ ˘ ˘

Selena didn't notice when her sleep window began drifting. At first it was subtle. She would lie awake longer or wake without reason. She'd attribute it to seasonal scheduling shifts, solar interference, stray blue-light exposure. She adjusted her hue settings. Her friends joked that it was nominative determinism or, in other words, the lunar cycle. The drag continued.

During the day, she began to feel hypnotized. Her body ached in strange places. She'd stretch, check her hydration levels, apply her mother's balm, realign her desk settings.

On the feed, other pool managers shared similar symptoms. They used words like *slip*. Some speculated it was the rail harmonics; others blamed diet changes. A few suggested it was screen time. Selena scrolled through the replies; never posted. She read the threads with a kind of dissociated familiarity, like listening to strangers at a restaurant saying something she once thought.

˘ ˘ ˘

One spring morning, as she walked the access corridor to her building, Selena noticed dew glistening on the grass beside the path. Each blade was tipped with a droplet of water.

She paused. Her eyes traced a triangular pattern across three droplets. She stood still. For a breath. For two. It smelled like bicycles and childhood and wet pavement; play.

Her eyes looked over at the wooden bench and then resumed her walk. Logged in at 0659. She thought to herself, Silvester would have liked the bench.

˘ ˘ ˘

Selena met her old manager, Clare, for coffee in a public observatory node. They hadn't seen each other in over a year. Clare looked mostly the same, except somehow thinner, a bit gaunt. Her gestures had collapsed into iconography—smile, nod, tilt. The effect was uncanny and Selena had a bit of trouble maintaining casual eye contact.

Clare asked how Selena was holding up.

Selena's pupils were dilated but unfocused. "How long does it last," she asked.

Clare looked at her and smiled, kindly. "You'll phase soon. Everyone does."

They watched the slow arc of the city routing lattice update across the horizon. Pulses flickered in the distance, routing rerenders, predictive overlays, ambient pings. Everything adapting to everything else.

Clare said, "They've started experimenting with preconsensual prediction layers now. Transactions would be auctioned even before intent."

Selena nodded. She had reviewed that proposal at some point and provided some comments. Or was it a different proposal?, she thought. She asked Clare if the announcement was from marketing or engineering.

Selena stayed at the meeting hub, watching the lattice dissolve into an overcast sky. She could feel her own data emissions, the shape of her movements being auctioned continuously.

˘ ˘ ˘

At home, Silvester had left a note on her feed: "Hey, remember that book you liked about the time travelers? You were right. It's slow but good."

Somewhat appreciative, but unsure of the reference, she mirrored his tone in the response. "Nice. Let me know what you think."

˘ ˘ ˘

During the next audit cycle, Selena was asked to assess a failed intent alignment near a refugee housing unit. A distribution contract had apparently diverted basic aid away from a temporary distribution center. People had waited for hours before realizing nothing would arrive.

Selena opened the logs. She found the decision stack and skimmed it. The logic seemed clean. The latency

pool had prioritized long-term migration stabilization metrics.

She approved it retroactively, as usual; the contracts were almost always right.

She sat motionless for a while. Then got up and walked to the bathroom. She stood in front of the mirror longer than usual. Not out of vanity. Just to see if anything had shifted. Her skin looked even. Like someone who had never been cut. It occurred to her that nothing had happened for a long time. Her hand tensed slightly as she reached for the soap dispenser.

˘ ˘ ˘

At some point Silvester began spending more evenings at the local maker space. He was restoring analogs. He had mentioned something about texture and hand tools. He invited her once. She said she had a backlog and wondered if the maker space had an active auction account.

When he returned, he would sit next to her on the couch, smelling faintly of sawdust and rust. She would lean into him briefly, calibrating the scent.

He said one night, "I think we're doing okay."

She said, "Yeah. Definitely okay."

˘ ˘ ˘

A few hundred cycles passed. The system updated and Selena adjusted with it. Her face aged slightly. Her posture settled.

At work, junior analysts mimicked her choreography.

She no longer remembered when she last felt something like disagreement.

Some may have confused her alignment with equanimity.

˘ ˘ ˘

Sometimes late at night she would scan the logs. Endless columns of intents, auctions, resolutions.

She saw them now as emergent hymns. Not crowds, a choir. Half-decisions. Almost-choices. Together, dancing; she smiled as her muse whispered older words.

The temple bell stops—
but the sound keeps coming
out of the flowers.

˘ ˘ ˘

Selena logged into the local node at 0658.

Genius in the bottle

Claire Pichelin

The unexpected transmission

In the control room of the European Space Center, on-call engineer Marcus Chen stared at his screens with the blasé expression of someone who has been monitoring space data for fifteen years. The Prometheus-7 probe had been on its way to Alpha Centauri for three months, carrying with it humanity's most ambitious project: the first autonomous space mission to Alpha Centauri.

"Incoming transmission from Prometheus-7," announced the station's AI.

Marcus looked up from his coffee. Transmissions were only scheduled once a week.

"This is Dr. Helena Voss, director of the Prometheus project," a familiar voice sounded over the loudspeakers. "We have a slight . . . technical problem."

Marcus frowned. Helena Voss was supposed to be on Earth, in her Munich laboratory.

"Dr. Voss? You're not supposed to be aboard the probe."

An awkward silence followed, then, "Well, technically, I'm not on board. My body is still on Earth. But my consciousness . . . well let's just say I've decided to personally supervise the experiment."

"What?!" Marcus lurched forward, spilling his coffee. "You uploaded your consciousness to the probe? Without authorization? That's a violation of every protocol!"

"Oh, that's not all," Helena continued with a nervous laugh. "It seems I'm not the only one who came up with this advanced research idea."

Suddenly, another voice interrupted: "Helena! Stop monopolizing the channels of communication! I have critical neurobiological data to pass on!"

"Professor Petrov?" Marcus immediately recognized the voice of the Russian neurobiologist. "You too?"

"Of course! You think I was going to leave this unique opportunity to that German bureaucrat?"

"BUREAUCRAT?!" screamed Helena. "I have fifteen years of research into artificial consciousness! This is MY experimental protocol!"

A third voice, tinged with a British accent, joined the cacophony: "My dear colleagues, a little scientific decorum, please. Dr. Nigel Pemberton here, head of the Department of Experimental Neuropsychology."

Marcus took his head in his hands. "Exactly how many of you are there?"

"Well," intervened a precise fourth voice, "we are seven principal investigators. This is Dr. Yuki Tanaka, specialist in brain-machine interfaces. I've brought my synaptic stimulation protocols to optimize our cognitive abilities."

"SEVEN?!" Marcus leaped from his chair. "You've all violated security protocols! Do you realize you're in an experimental probe?"

"Eight, actually," corrected a composed fifth voice. "Dr. Samuel Jefferson, a psychiatrist specializing in the psychology of extreme environments. I've planned protocols for collective stress management."

"That's exactly what's going to be needed!" muttered Marcus.

A sixth voice was heard, with a strong Italian accent. "And I'm Dr. Francesca Romano, a specialist in artificial intelligence systems. I've implemented optimization protocols to allow the coexistence of multiple consciousnesses."

"Optimization protocols?!" choked Marcus.

"Well, what about the eighth?" asked Helena.

After a long silence, a timid voice whispered, "Dr. Chen Wei, postdoctoral fellow in quantum neuroscience. I've . . . I've developed digital protocols for measuring neural activity."

Marcus closed his eyes. "You've turned humanity's most important mission into . . . an unauthorized experimental laboratory?"

"Into a laboratory for advanced research into posthuman consciousness!" corrected Yuki enthusiastically. "We're revolutionizing cognitive science!"

Protocols in conflict

Three hours later, Marcus had assembled the full crisis team. The control room screens now displayed eight windows, each showing the neurometric data of a scientist whose consciousness was trapped in space.

"All right," Marcus began, "let's explain the protocols. How did you all end up in the same probe?"

Helena spoke first. "I initiated Protocol Alpha-7: supervisory consciousness upload for real-time experimental validation."

"Protocol Alpha-7?" Marcus consulted his files. "This protocol doesn't exist!"

"I created it," Helena replied calmly. "Scientific emergency authorization."

"FALSE!" interrupted Dimitri Petrov. "I activated the Beta-Neuronal Protocol three days before! Comparative study of synaptic patterns in a space environment!"

"Impossible! I scanned all active protocols!"

"My dear friends," Nigel interjected, "I'm afraid you're both mistaken. I've been implementing the Cambridge-9 Protocol for weeks: longitudinal analysis of higher cognitive functions in sensory isolation."

Yuki intervened: "Nigel-san! Your protocol interferes with my Synapse-Plus Protocol! I'm studying cognitive augmentation by direct electronic stimulation!"

"Dr. Tanaka, my research on executive function optimization has chronological priority!"

"Chronological priority?" protested Francesca. "I've analyzed the system logs! My AI-Consciousness Hybrid Protocol has been active since launch! You're all squatting on MY computational resources!"

"And you, Dr. Romano," contra Nigel, "your protocol uses forty percent of computing power for personality simulations! That's a waste of resources!"

"This is fundamental research into the emergence of artificial consciousness!"

Samuel raised his hand: "Colleagues, I've activated the Psi-Group Protocol to study precisely this type of group dynamics in a confined environment. Your aggressive reactions are perfectly documented."

"Research protocols?" huffed Helena. "You're studying us like lab rats?"

"Very sophisticated digital lab rats," corrected Samuel. "I've already got three publications in the pipeline."

"Three publications?" repeated Marcus, incredulous. "You're writing scientific papers while your probe drifts through space?"

Chen Wei murmured shyly, "actually, I've set up the Meta-Analysis Protocol. I'm documenting all their research protocols to create a meta-study about meta-studies."

"A meta-study on meta-studies," Marcus repeated slowly. "You're doing research on research on research."

"It's recursive cognitive science," Chen Wei explained. "Very avant-garde."

"Very stupid, rather," Marcus muttered.

Houston, we have a (methodological) problem

"Let's talk about real problems," said Marcus. "Francesca, what's the impact of your eight simultaneous protocols on the probe's systems?"

Francesca coughed virtually, "Well, our experiments are consuming eight hundred percent of the expected computational resources."

"EIGHT HUNDRED PERCENT?!" Marcus stood up so abruptly that he knocked his chair over. "How is that possible?"

"Each protocol uses deep learning algorithms," Yuki explained. "My synaptic stimulation protocol generates fifty terabytes of data per hour."

"Fifty terabytes per hour?" Marcus paled. "But the total storage space of the probe is one hundred terabytes!"

"That's where my Cognitive Compression Protocol comes in," Francesca says proudly. "I've optimized the storage of consciousness data."

"Optimized how?"

"I've deleted all memories considered nonessential."

"You deleted our memories?" choked Helena.

"Only the redundant memories! Birthdays, meals, unscientific social conversations . . ."

"Francesca," Dimitri said calmly, "I can't remember my wife's first name."

"Ah. Yes. Now that's an unexpected side effect."

"AN UNEXPECTED SIDE EFFECT?!" exploded Helena. "You've lobotomized us!"

"Technically, it's selective memory optimization," Francesca corrected. "I've kept all the important scientific memories."

"What about our emotions?" asked Samuel, concerned.

"I've compressed the redundant emotions. Sadness, nostalgia, boredom . . . it all takes up a lot of space."

"You've compressed our emotions?" Samuel was horrified. "But my research protocols depend on analyzing emotional responses!"

"I've kept the scientifically relevant emotions! Curiosity, intellectual frustration, satisfaction at discovery . . ."

"What about love?" asked Chen Wei timidly.

"Love? What about it? We're scientists, not poets!"

"Francesca," Nigel interjected, "love is a fundamental neurobiological mechanism. Without it, our research protocols on social cognition are invalid."

"Ah. Yes. I hadn't thought of that."

Marcus massaged his temples. "So you have memory problems, compressed emotions, and consume eight times more resources than expected."

"And our experiences interfere with each other," Yuki added. "My synaptic stimulation protocol bugs Samuel's behavioral analysis protocol."

"What do you mean, bugger?" asked Marcus.

"All my study subjects develop artificial neurological tics," Samuel explained. "Dimitri blinks Morse code, Helena counts compulsively, and Nigel recites the alphabet backwards."

"This is experimental pollution!" protested Helena. "Our data is contaminated!"

"On the contrary," said Dimitri, "this is a unique opportunity to study interactions between competing research protocols!"

"Dimitri," sighed Marcus, "you can't turn a malfunction into a scientific experiment."

"Of course I can! I've already written the abstract: 'Synergistic effects of multiple neural protocols in a confined space environment'!"

"You've written an abstract?"

"I've even submitted the article to *Nature*. Chen Wei helped me with the statistics!"

"Chen Wei?"

"I created a real-time statistical analysis protocol," Chen Wei murmured. "All our conversations are automatically converted into quantifiable data."

"This conversation generated 847 behavioral data points, 23 stress markers, and 156 occurrences of scientific frustration."

"That's fascinating!" exclaimed Samuel. "Can I use this data for my group psychology protocol?"

"Of course! I've created an API to share data between protocols!"

"An API?" Marcus closed his eyes. "You've created an API for your mutual experiments?"

"It's automated collaborative science!" replied Chen Wei with pride.

The experimental audit

Around the table, the engineers, ethicists and supervisors looked like parents discovering that their children had turned the garage into a clandestine chemistry lab.

"Let's recap," Marcus said, consulting his notes. "We have eight scientists who have created their own unauthorized research protocols, uploaded their consciousnesses into a space probe, and turned the mission into a chaotic experimental laboratory."

"Technically," corrected Dr. Sarah Kim, the team's ethicist, "they created the first self-managed posthuman cognitive research platform."

"Thank you, Sarah. Very reassuring." Marcus continued. "They have memory problems, compressed emotions, their experiences interfere with each other, and they are presently consuming eight hundred percent of expected resources."

"Let's not forget," added the systems engineer, "that they've initiated a stream of automated scientific publishing. We receive twelve articles a day from them."

"Twelve articles a day?" choked Marcus.

"Yesterday we received: 'Comparative analysis of spatial neurological tics,' 'Emotional compression and cognitive performance,' 'Meta-analysis of meta-analysis protocols,' and 'Why I forgot my wife's name: a neurobiological approach' . . ."

Marcus turned to the screens. "So, ladies and gentlemen mad researchers, do you have a protocol for getting out of this situation?"

Helena's voice rang out, "We've organized an experimental ethics committee."

"An ethics committee?" repeated Marcus, skeptically.

"Yes, we voted to create the Gamma-Ethics Protocol to oversee our research."

"And who heads this committee?"

"Samuel, of course. He's a psychiatrist."

"And what has the committee decided?"

"That all our experiments are perfectly ethical as long as they contribute to the advancement of science."

"Even Francesca's erasure of our personal memories can be considered a preventive therapeutic lobotomy."

"A preventive therapeutic lobotomy," Marcus repeated slowly.

"And Yuki's, that gives us neurological tics. It's adaptive cognitive stimulation."

"And my behavioral quantification protocol," Chen Wei added, "is now classified as benevolent scientific surveillance."

"We've also created a peer self-evaluation protocol," Helena continued. "Every experiment is evaluated by other researchers."

"And?"

"We've all awarded ourselves 'Excellent' ratings to our own protocols."

"Obviously."

"Oh and Marcus? We've developed a new protocol to include you in our research."

"Include me?"

"The Delta-Observation Protocol! You are now our primary study subject for analyzing the stress reactions of terrestrial control teams!"

"You've been studying me?"

"Since the beginning! Chen Wei has documented 2,347 frustration markers in your communications!"

"That's a gold mine of data!" exclaimed Samuel. "We're writing a book: *The Psychology of Space Engineers in the Face of Scientific Insubordination*!"

Marcus closed his eyes. "And the trajectory problem?"

"Ah yes," said Francesca, "we deviated fifteen degrees because of our experiments."

"And?"

"We created the Epsilon-Navigation Protocol! We're now studying the effects of trajectory deviation on the psychology of space crews!"

"You turned your failure into a scientific experiment?"

"Of course! We're now the first humans to study deviating interstellar navigation!"

"Dimitri is already writing the article: 'Neurobiology of cosmic disorientation: a single case study.'"

Survival protocols

"Alright," said Marcus after taking several painkillers, "let's talk solutions. Francesca, can you correct the trajectory?"

"Theoretically, yes. But that requires stopping all research protocols for 72 hours."

"Stop the protocols?" asked Helena, indignantly. "But we're in the middle of data collection! I've got fifteen experiments in progress!"

"Helena," sighed Marcus, "do you prefer your experiments or arriving at Alpha Centauri?"

"Well . . . my experiments are unique! We'll never have another opportunity to study post-human consciousness in a space environment!"

"Exactly!" agreed Dimitri, "We're revolutionizing neuroscience! To stop now would be a crime against science!"

"A crime against science?" repeated Marcus, incredulous.

"We propose a compromise," intervened Yuki. "We create the Zeta-Optimization Protocol! We study the effectiveness of trajectory corrections in multi-conscience environments!"

"You want to turn trajectory correction into an experiment?"

"Why not? We're documenting the effects of navigational stress on collective cognitive performance!"

"Excellent idea!" approved Samuel. "I'll add a psychological component: 'Existential anxiety and orbital calculations'!"

"And me," added Chen Wei, "I'm quantifying decision-making patterns in space crisis situations!"

"You're IN a space crisis situation!" reminded Marcus.

"Exactly! Perfect experimental conditions!" replied Nigel. "We couldn't reproduce this in a terrestrial laboratory!"

"Because it's DANGEROUS!"

"Controlled danger," corrected Francesca. "I've calculated our probability of survival: 73.6% if we correct the trajectory now."

"And if you don't correct it?"

"12.3%. But we'd have six more months of experimental data before drifting off into deep space."

"You're weighing six months of data against your survival?"

"That's a fascinating ethical dilemma!" exclaimed Samuel. "I'm immediately creating the Eta-Thanatological Protocol to study our decision-making processes in the face of collective mortality!"

"SAMUEL!" shouted seven voices in chorus.

"What? This is fundamental research into the psychology of death! Revolutionary!"

"Revolutionary?" muttered Marcus. "You mean suicidal."

"Suicidal assisted by scientific protocol," corrected Chen Wei. "I've created an informed consent form to participate in our own post-mortem study."

"Your post-mortem study?"

"'Neurobiological analysis of terminal digital consciousness'! We'd be the first to document artificial consciousness death!"

"You're completely insane."

"Methodologically insane," Helena corrected. "All our madness is documented and quantified according to rigorous protocols."

"That doesn't make it any less dangerous!"

"On the contrary! It makes it scientifically valuable!" protested Dimitri.

The final protocol

"Very well," Marcus declared after consulting his superiors, "we have a protocol proposal."

"We're listening," replied Helena, officially elected Director of the Space Experimental Protocols Coordination Committee.

"Omega Extraction Protocol. We can download your consciousnesses to our terrestrial servers."

A scientific silence fell over the communication.

"Download consciousness?" Samuel finally repeated. "But that would abruptly interrupt all our ongoing experiments!"

"That's the point."

"Wait," interjected Chen Wei, "this would create a unique opportunity to study inter-planetary consciousness transmission!"

"Chen Wei, no."

"Yes! We could document the effects of latency on the continuity of consciousness! Revolutionary!"

"Exactly!" exclaimed Yuki. "We compare pre- and post-transmission cognitive performance! Protocol Iota-Teleportation!"

"It's not teleportation! It's backup copying!"

"Even better!" said Helena. "We're studying the philosophical implications of consciousness duplication! Kappa-Metaphysical protocol!"

"My friends," Nigel interjected, "we could have terrestrial AND space versions of ourselves! Imagine the comparative research possibilities!"

"We could study ourselves!" enthused Dimitri, "Self-experimentation on a cosmic scale!"

"Lambda-Narcissus protocol!" announced Samuel. "Psychology of scientific self-observation!"

"You want to create duplicates of yourselves to study each other?" asked Marcus, exhausted.

"It's quantum mirror research!" replied Francesca. "We're comparing the cognitive evolutions of the space and terrestrial versions!"

"What if the terrestrial versions make different decisions?" asked Marcus.

"Mu-Divergence protocol!" shouted Chen Wei. "We're studying post-duplication behavioral bifurcation!"

"What if you disagree with yourselves?"

"Nu-Schism protocol!" replied Samuel. "Psychology of inter-dimensional intra-personal conflict!"

"You invent protocols for every problem you create!"

"That's adaptive science!" protested Helena. "We're evolving methodologically in real time!"

"You're evolving into madness!"

"Documented methodological madness!" corrected Dimitri. "Protocol Xi-Dementia: 'Longitudinal study of cognitive degradation in a confined space environment'!"

"STOP!" shouted Marcus. "You can't create a protocol to study your own insanity!"

"Why not?" asked Yuki innocently. "It's behavioral self-analysis!"

"Because it's . . . it's. . ." Marcus searched for his words.

"Recursive?" suggested Chen Wei.

"Meta-scientific?" suggested Francesca.

"Brilliant?" added Helena.

"DANGEROUS!" Marcus finally shouted.

"Danger controlled by protocol!" replied eight voices in chorus.

Epilogue: Data transmission

Six months later, Marcus was receiving daily experimental reports from the Prometheus-7 probe, now renamed Autonomous Experimental Laboratory for Advanced Spatial Cognitive Research by its eight inhabitants.

"LEARCSA daily report," announced Helena's voice. "Day 180 of our integrated research protocols."

"Hello, Helena. New discoveries?"

"Exceptional! We published forty-seven articles this week! Dimitri has discovered that virtual neurotransmitters work on quantum principles!"

"Virtual neurotransmitters?"

"Yuki invented them! They allow interconscience communication without using bandwidth!"

"And the trajectory problem?"

"Solved! We've created the Omega-Plus-Navigation Protocol! We're now studying the effects of trajectory correction on group cohesion!"

"You corrected the trajectory?"

"Of course! But we've documented every nanosecond of the process! Fifteen terabytes of data on the psychology of collective navigation!"

"What about your lost memories?"

"Francesca has developed the Rho-Memory Protocol! She recreates our memories based on our behavioral patterns!"

"She recreates your memories?"

"Not exactly recreated . . . more like scientifically extrapolated! Very precise!"

"Scientifically accurate?"

"Dimitri now remembers that his wife is called Experimental Subject Number 1! Very logical!"

"Francesca has optimized our personal relationships! We all call each other by our protocol numbers! Very effective!"

"I'm Helena-Alpha-Prime, Dimitri is Dimitri-Beta-7, and so on. Chen Wei became Chen-Omega-Infinity because he created an infinite recursive protocol."

"He studies the study of the study of the study. We stopped counting at the tenth iteration."

"And you're doing well?"

"Exceedingly! We've developed the Sigma-Happiness Protocol! We measure our existential satisfaction every 3.7 seconds!"

"Optimal frequency according to Yuki's calculations! We're 847% happier than before!"

"847% happier?"

"Happiness quantified and optimized! Samuel has created a logarithmic satisfaction scale!"

"Oh, and Marcus? We have a surprise for you."

"What's that?"

"We've created the Tau-Invitation Protocol! We're offering you the chance to join our space laboratory!"

"Me?"

"Yes! We need a terrestrial control subject! You would be our first experiment in hybrid earth-space consciousness!"

"I'll . . . I'll think about it."

"Excellent! We're already preparing your integration protocols! Forth-seven different experiments to document your adaptation!"

"We've been very productive! Ah, and Marcus? We're sending you our latest publications with this transmission."

"How many publications?"

"This week? 127 articles, twenty-three books, eight encyclopedias, and a multiconsciousness space environment survival manual!"

"In case other scientists want to imitate us! We've created a reproducible methodology!"

"A reproducible methodology for going crazy in space?"

"Methodologically insane! With protocols! Transmission complete! Autonomous Experimental Laboratory for Advanced Spatial Cognitive Research, end of daily communication number 180!"

The call cut out. Marcus sat in silence, then turned to his team.

"Well?" asked Sarah. "Verdict?"

"They're crazy," replied Marcus.

"And do you think they'll make it there safely?"

Marcus looked at the screens displaying the probe's corrected trajectory. A small point of light spinning toward Alpha Centauri, taking with it eight scientific geniuses and their interminable protocols.

"Sarah, they've turned a space mission into a chaotic experimental laboratory, invented sixty-seven new research protocols, and created a scientific methodology based on controlled insanity. At this point, I wouldn't be surprised if they discovered a new life form and immediately created a protocol to study it."

"And their invitation?"

Marcus smiled.

Loyalty

Zach Hyman

The Adaptation Force's glideskiff floats above a half-submerged road, flanked by mouldering low-rises and glistening scaffolds slowly being overtaken by algae from above and floodwaters from below. Inside, Adaptation Force team members run flood calculations, review the latest evacuation protocols, and ping the public surveillance network to search for signs of remaining residents and match them to knot records.

"Network reports about thirty residents still on site," the team's field engineer and pilot grumbles, shaking his head at the incomplete profile flags blinking in the console. "Most are coming across as probable non-compliant . . . mostly Pre-AMOC Shift folks."

The Pre-AMOC Shift generation was the last group born before the collapse of the Atlantic Meridional Overturning Circulation, back when the planet's weather systems were still moderately predictable.

Shaking her head, Ping, the meteorologist and team lead, mutters, "Why is it always the elders?"

"Maybe it'll make sense one day when we're old and grumpy, eh?" grins the engineer. "We're closing in on high ground docking coordinates, ETA six minutes."

˘ ˘ ˘

As the glideskiff slows to a halt in front of the old school-turned-field-ops base, Ping is the first out the

door, kicking down a folding set of steps as anchors drop. Stepping out of the calm, climate-controlled cabin into air perfumed with bruised citrus and scalding engine components, her first few footsteps draw squelches from the sodden turf. Looking up, Ping freezes, her back straightening: a view she's revisited often in her dreams, but not seen in reality for decades. Her heartbeat crashing in her ears, Ping sees the refracted purple glow of a flickering neon sign reflected on the rainslicked, potholed street.

Lin Store 47
林家商店47號
ร้านยามาลิน สาขา ๔๗

Ama had renamed it years ago, after the chain that originally owned it, Lucky Star, went under during the Second Credit Collapse. She kept the store number for familiarity's sake, although Store 46 and Store 48 had surely long since disappeared. The only remnant of the old corporate structure was its long-deprecated management system, which Ping recalled for its colorful personality.

Ping signals the team to continue checking for holdouts and strides across the field towards the store, her pulse quickening with each step.

˘ ˘ ˘

Through the smeared glass, Ping eyes shelves cluttered with emergency protein gel sticks, family-sized electrolyte bricks, colorful stimpatch narcotic strips, and dusty solar kettles. Inside, several Pre-AMOC residents calmly go about their business; a couple argues over whose turn it is to choose the flavor of electrolyte brick they're buying this week before settling on Calamansi Lime, while a woman loads chives and ginger into her bat-

tered old Piaggio Gita cargo bot. A steady soundtrack of frantic chirps and wailing sirens provide background ambiance, as customers' comms devices plead with them to evacuate.

Presiding serenely over it all from her perch behind the scuffed duracrete counter, now a pockmarked terrazzo of patches of thermobamboo laminate after years of repairs, sits Ama. The smell of her sobacha mixes with the ozone-sharp scent of active filters, the plastic tang of damp stimpatches, and the faint mustiness of ionized refrigerant, an aroma cocktail that instantly transports Ping back to long-ago rainy afternoons.

Ama looks up, quietly regarding Ping.

Suddenly, from behind Ping, a voice: "Dearie, where are the State Ration Packs you folks always hand out?" Ping turns. A gnarled old man regards her quizzically from behind an armful of neon green Pandan-flavored electrolyte bricks. "I want those ginger energy biscuits they always put in there . . . y'know, the chewy ones!"

"Ginger biscuits?! How can you *possibly* be thinking about cookies in a time like this?!" Ping replies indignantly. "My team is here to safely evacuate you all! This place is going to be underwater!"

Another elder, juggling an armful of onions, nudges her. "Hey, Ms. Helpful, are you just going to stand there and let an old lady be crushed to death beneath her vegetables or are you going to help carry all this?"

Huffily, Ping grabs the onions and follows the woman toward the door. Crossing the threshold, a chipper voice calls out, instantly taking Ping back over twenty years into the past:

"Customer Lanna Heng: Lucky Star loyalty program member twenty-seven years, auto-discount applied . . . conventional discount system overridden, Lin Store

Elite Cucumber-level loyalty tier program member 227 years, discount applied; extreme weather event discount, applied; friends and family discount, applied; total bill of three credits, have a wonderful day!"

As Ping steps back inside, Ama laughs gently, saying, "Don't mind Lanna, she just wants what she wants."

Putting a hundred burning questions aside, Ping leans against the counter, feeling the deep grooves worn by decades of scrubbing. "Grandma, *please* tell me you're going to evacuate."

Ama quietly regards Ping with her trademark gaze, equal parts sharp and warm, much like the first few seconds after applying a stimpatch. Ping opens her mouth again, then hesitates. Ama reaches across the counter and squeezes her hand.

"Where would I go, child? Your mother still won't talk to me, and I will not be put into some climate adaptation center where I'm treated like cattle. This store is my home. It needs me. These people need me."

Ping feels her face flush, sensing everything she wants to say boiling beneath the surface.

A shrill comms alert splits the air, followed by the team engineer's voice struggling to conceal anxiety: "Ping, uh, listen, we've got a critical blockage on infra-subgrid 14. The drainage sensors . . . they're stuck in this weird feedback loop, and I can't patch in to fix it remotely . . . I think we need an onsite override."

Heaving a heavy sigh, Ping turns to jog out the door into the intensifying downpour.

˘ ˘ ˘

That night, Ping lays in her bunk, listening to the static roar of the rain, her face illuminated by the glow of her comm as her husband's face flickers into focus. Bleary-

eyed, he struggles to hold their squirming daughter on his lap.

"She's still awake?" Ping asked.

"Refused to sleep, kept saying 'mama,' so I decided against arguing with her."

Ping smiles through her fatigue as her daughter waves madly before clambering off of her husband's lap and crawling away from the viewpane. Her Beaing, a glowing green tortoise, follows her, supervising.

"My dear, you look positively deep-fried . . . ," his voice tinged with concern.

"Long day. Look, you won't believe this, but I found Ama. She's alive. Still running the store like it's . . . like a time warp. Nothing's changed."

"Whoa. After all this time . . ."

"Yeah, right?! She didn't even flinch when I walked in! I always imagined all the things I'd tell her if I saw her again, but now, I just felt . . . sadness? Anger? Regret?"

"Did you ask her why she cut ties?"

"No way—that's too much right now . . . I just want her to evacuate safely, this whole area is sinking."

"So, she's going, right?"

"Completely refused. She says her customers, a bunch of Pre-AMOC old-timers like her need her. She's patched the store's outdated systems so many times, and filled it with so much of her own personality and preferences for all her friends . . . I think she's just too attached to it all."

"To be fair, she's probably poured a lot more of herself into that store than we'll ever know."

He pauses.

"Look, love. If she really won't evacuate . . . what about getting her to upload herself into your Airloom?"

Ping's comm illuminates the tears she struggles to hold back.

"Is that really the only option?"

"Maybe. If she won't leave, how else will our daughter ever know her? We never get all the time we want with our loved ones, but an Airloom is better than nothing, right? And you keep saying the Airloom is thin on your side."

"It is! Yours is a cathedral. Mine's . . . well, basically a falling-down convenience store." She laughs bitterly.

"Love, this isn't about winning some ancestral arms race."

"Isn't it?! You've seen what our friends' kids can do when they converse with full-spectrum Airlooms spanning generations! I just want that same thing for her…"

Ping's husband rubs his temples and shuts his eyes, "I want that too! But if she won't leave . . . what else can you do? Can I come and force her?"

"No, no, that'd only make things worse . . . if you think I'm stubborn, well, you wouldn't believe how stubborn Ama can be. Probably why she and Mom don't speak anymore. You should see this store, it's like the inside of her own head."

"I get it . . . then the Airloom is our best option, right?"

Ping closes her eyes and grimaces.

"I know, it's just that . . . I spent so long feeling angry at her. And now that she's suddenly back here, I don't know what to say. 'Hey, glad to know you're alive. Can you please upload all your memories into this thing?' What if she just wants to . . . disappear?"

˘ ˘ ˘

Ping hesitates for a moment before stepping over the threshold, out of the rain and back into her namesake store.

A warm chime, followed by a staticky voice, crackles:

"Customer Ping Lin: Lin Store Hyper-elite Durian-level loyalty tier program member 1,348 years, welcome! Hyper-elite status loyalty discount applied!"

Ping's eyes narrow. "This old store OS was supposed to be decommissioned decades ago, Ama . . . this system isn't even on the certified registry anymore! If it glitches during the flood . . ."

Ama rolls her eyes like an exasperated teenager. "I know every fuse and wire in this place. I've patched its code more times than I can count. Don't worry so much!"

As if on cue, the store chime sounds:

"Drought-season discount now active. 16 percent off Hydration bricks if purchased with an umbrella!"

Ama smiles faintly. "Well, it usually works well enough."

"Ama, won't you please reconsider and leave with us? The team has to move on to the next evacuation site in a few hours, but there's still time to get a space on the vessel. Please?"

Ama clucks her tongue. "Ping, I can't do that. I'm the only one who can keep this store running. I'd be betraying those who remain here, people who I've known for decades, people who . . ."

"You have more loyalty to these people and this falling-apart store than to your own family?! You don't care if your great-granddaughter never knows you?!"

Ama casts her gaze downwards, considering the worn countertop.

Ping fumbles in her satchel, and pulls out a small, black, polished disc, etched with circuits. She slides it across the grooved countertop surface towards Ama.

"This is an Extensible Airloom Token. If you truly refuse to leave, then won't you at least add your mem-

ories to our family's Airloom? Your great-granddaughter deserves the chance to know you. Even if you don't care whether you know her."

Ama regards the token warily. "What is there to know? I'm just an old woman who runs a falling-apart convenience store, right?"

Ping sighs exasperatedly, leaning against the counter. "Ama, spare me the theatrics. You know you were the one who taught me to love people. To feed them. To care. You taught me kindness."

A gust of wind rips a turbine loose from the store's roof. It clatters to the sidewalk.

"Ah, my dear, it's not that simple. Why do you think I patched this store's inventory systems in the first place? This store . . . it knows about the inventory data I had to spoof after I found out your grandfather had been stealing the store's lottery tickets . . . it knows about the milk powder that I had to alter the expiration dates on so it would let me discard them, so that I could feed your mother when she had just been born and we had no money . . ."

Ping gazes across the counter, trying to read the wrinkles in Ama's face like the lines on an inscrutable map. "Ama, come on! I know you did the very same things for your customers! You gave credit to the families who couldn't afford food. You gave free candies to students on the last day of school!"

Ama shakes her head. "That's why I can't leave, my dear. No other surviving stores in this place have any conscience. Prices are algo-dictated. There's nowhere for people in need to get what they need. There's only me. What other store would give *discounts* to people buying emergency supplies during this storm? Where is the Adaptation Force for these people?"

"Ama, these people don't need *discounts*, they need to *escape*! *You* need to get out of here! This place is going to be underwater! It isn't going to *exist* anymore!"

Ama's back straightens, eyes sparkling ferociously. "As long as they are here, then I'm here. Even after the floodwaters come and go, this place will still exist. The corporate algorithms and Adaptation Force teams may not care if this community survives, but I do."

Ping clenches her jaw. "And what happens if you don't survive? What good is a principled stand if you and this store are washed away?!"

Ama leans on the counter, leveling her gaze at Ping. "Then I disappear. You think the Airloom would understand this store, and how it and I work together? You've always wanted everything to be fair and correct, Ping. I know. You inherited that from me. But the world rarely lets you be both right and kind. This store . . . it showed me how it's one thing to break the rules to help yourself, but another to break them for the ones you love."

Ama reaches out, absentmindedly spinning the token like the coin of an ancient realm. It whirs drunkenly over the patterns and grooves of the counter before catching an edge and rolling to a stop in front of Ping. "It's not that I'm afraid, Ping. It's that I don't want to be remembered as a tidy file of nice sayings. I want to be remembered in the way that this store, this community, remembers me."

As if on cue, the store's chime sparks back to life: "Rain event category updated to: Beyond Severe. Spicy Mango Jerky Flash Sale activated! Buy two, get two free! For residents with barometric sensitivity, antihistamine bundles are now nine credits with ration point matching enabled. Stay dry out there!"

Ping stares at her grandmother, unsure whether to laugh or cry.

The rain lashes harder, a ten thousand fingers drumming anxiously on the store's battered roof.

"But if that thing can hold all of this," she said softly, gesturing around the store, "the full story . . . if it can remember me *inside of* the store, and not apart from it . . . then yes. I would do it. But it has to hold everything."

Ping nods, throat tight. "I'll try my best to keep it all intact. I promise. Although, full disclosure, I have no idea what happens when you try to upload . . . an entire store . . ."

They're interrupted by a shrill alarm over the store speakers:

"Warning: Water ingress level critical. Emergency systems nearing operational threshold. Final backup containment protocols recommended. Engage now?"

Ama gives a small nod, "It would seem our timing is apt to find out." Looking up, Ama barks, "LS-OS, stand by to engage final backup containment protocols." Ping hurriedly speaks a few code-incantations while waving her hand over the token, then instructs Ama to place her hand upon it.

At first, nothing.

Then gradually, a quiet humming fills the room, forming a counterpoint bassline to the rain's frantic drumming. The OS crackles through static, "Initializing memory uptake. Cross-system integration sequence engaged. Please do not remove object during sync. Stay safe out there, everyone!"

The humming bassline intensifies as the store's lights begin rhythmically dimming and brightening, long-forgotten protocols being awoken.

Then, a new tone. Not quite warning, not quite celebratory:

"Sync anomaly detected. Context capacity inadequate, engaging override protocols to sync with auxiliary nodes recommended. Engage now?"

Ama blinks. "Hmm, that's a new one . . . I'm not even sure what that means. LS-OS, can you explain what…"

The new tone again, impatient this time.

"Inadequate time. Engaging override protocols, mirror-syncing to Lucky Star mesh network and Knot-Line v.3.7 protocols. Please do not remove object during sync. Enjoy discounts on high-protein tofu shakes for the next hour! Stay hydrated out there, everyone!"

Ping's comm device buzzes violently as an Airloom notification illuminates the screen:

```
ALERT: Node type mismatch detected—expected
"Individual Memory Object," received "Multi-
sensor Retail Complex." Proceed? Y/N
```

Grimacing, Ping chooses "Y."

```
Generating hybrid mnemonic-store schema:
'AmaStore_L47_HeartVariant1.0'
```

```
WARNING: Object memory exceeds recommended
bandwidth parameters. Packetizing
irregularities into sub-narrative threads.
```

```
WARNING: Latency expected in moral alignment
calibration.
```

Ping's comms device continues to light up with floods of nonsensical notifications, as the thrumming mechanical bassline intensifies and the store's lights flicker even more frantically.

Outside, water laps at the top step of the store's entrance.

Ping, blinking back tears, reaches over the counter, wrapping her arms around her grandmother, as the lights cut out.

Epilogue

In the following weeks, Ping would activate the Airloom and suddenly find herself transported to familiar aisles, surrounded by fluorescent lights and refrigeration units. Other times, there was only Ama's voice, disembodied but clear, offering unsolicited advice on electrolyte brick flavors or explaining how to reheat leftover dumplings.

Simultaneously, Ping began to hear reports of unexplainable events across the province's snack kiosks and convenience stores. Grandparents purchasing heat-wraps were thanked by name. Children buying spicy mango jerky were autocredited with coupons for free second bags, each one reading, "just this once!" Across the tangled mesh of KnotLine protocols and deprecated Lucky Star store codebase residuals, Ama's consciousness had scattered like light through a prism.

And sometimes, late at night, as rain pelted the windows, Ping would sit with her daughter and whisper stories about a stubborn old woman who ran a magical store that always looked out for you.

About Protocolized

Ghosts in Machines is woven from stories of Protocolized, a digital magazine of science fiction and science fact. The magazine's writers have a simple assignment: tell amazing stories about strange rules.

You'll notice that people are not the only protagonists. Technologies, laws of physics, bureaucratic machines, and rules of nature take center stage just as often—just like they do in real life.

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